

Nomenclature and Instrumentation

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Basic to any science is a language understood by the members of the community of the discipline. This chapter is devoted to a review of the language of operative dentistry and to the basic instrumentation of that discipline. *Operative dentistry* may be defined as that part of restorative dentistry involving assessment, prevention, and treatment of diseases and defects of the hard tissues of individual teeth to maintain or restore functional, physiologic, and esthetic integrity and health.

Nomenclature and Classification of Caries Lesions and Cavity Preparations

Systems for naming and numbering teeth

Each tooth may be identified by its location in the mouth and by its individual name. Examples include the maxillary right central incisor and the mandibular left second premolar. Areas of the mouth are referred to by *arch* (maxillary or upper and mandibular or lower) and by the side of the patient's midline (left and right) (Fig 7-1). Each arch is divided in half at the midline, forming four *quadrants* (maxillary

right and left quadrants and mandibular right and left quadrants). In addition, each tooth is identified as *primary* or *permanent*. Finally, the *individual name* of the tooth, eg, molar or central incisor, completes the identification. Examples of complete tooth names are mandibular left permanent first molar and maxillary right primary canine.

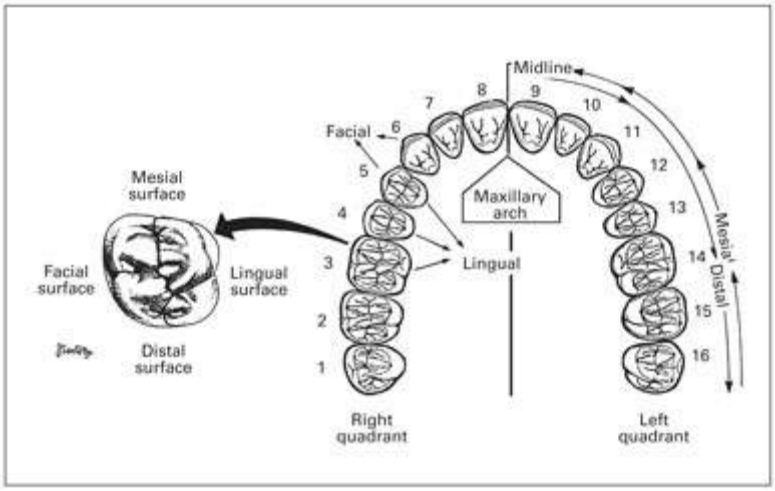


Fig 7-1 Nomenclature of directions and tooth surfaces.

Because their names are cumbersome, teeth are frequently referred to by number. The tooth-numbering systems primarily used today are the Universal system and the Fédération Dentaire Internationale (FDI) system (Fig 7-2). In the Universal system, the numbering begins with the maxillary (upper) right third molar (tooth 1), proceeds around the arch to the maxillary left third molar (tooth 16), then continues with the mandibular (lower) left third molar (tooth 17) and around the mandibular arch to the mandibular right third molar (tooth 32).

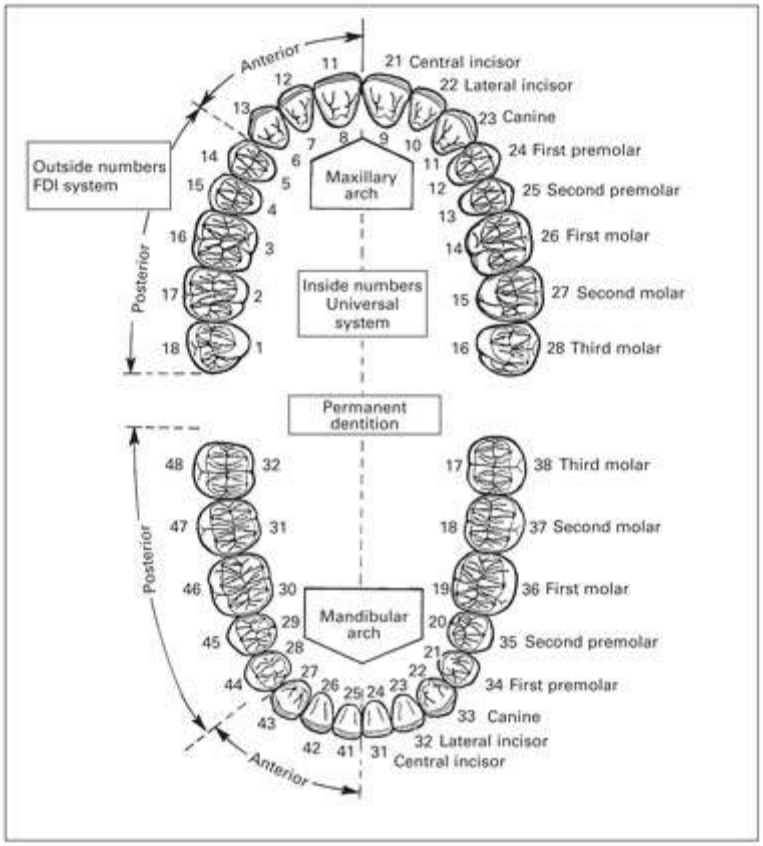


Fig 7-2 Tooth-numbering systems and nomenclature.

In the FDI system, the first digit of the tooth number represents a quadrant (1, maxillary right; 2, maxillary left; 3, mandibular left; and 4, mandibular right). The second digit represents the tooth (1, central incisor, regardless of the arch or quadrant; 2, lateral incisor; 3, canine; and so on to 8, third molar). The maxillary left first premolar would be identified as tooth 24; the mandibular right second molar would be identified as tooth 47.

Incisors and canines are referred to as *anterior teeth*, regardless of the arch; premolars and molars are referred to as *posterior teeth*.

In addition to quadrants, the mouth may also be divided into *sextants*, or sixths. There are three sextants in each arch, with divisions between the canines and first premolars—the maxillary right, anterior, and left sextants and the mandibular right, anterior, and left sextants.

Nomenclature of tooth surfaces and cavity preparations

The surfaces of the teeth are identified by their locations. Any surface or movement toward the midline of the arch is referred to as *mesial* (see Fig 7-1). A surface or

movement away from the midline is *distal*. Surfaces and movements toward the tongue are termed *lingual* (or *palatal* in the maxilla); those that are in the direction of the cheek or lips are termed *facial*. For the anterior teeth, facial may be referred to as *labial* (toward the lips); for posterior teeth, facial may be referred to as *buccal* (toward the cheek).[1]

On any tooth, *gingival* refers to an area or movement toward the gingiva (Figs 7-3 and 7-4). A distinction is made, however, between the chewing surfaces of posterior teeth, which are called *occlusal* (see Fig 7-4) and the biting edges of anterior teeth, which are called *incisal* (see Fig 7-3). A *proximal* surface is one that faces an adjacent tooth; it may be further identified as mesial or distal.[1] The contact between two adjacent teeth is referred to as the *interproximal contact*.

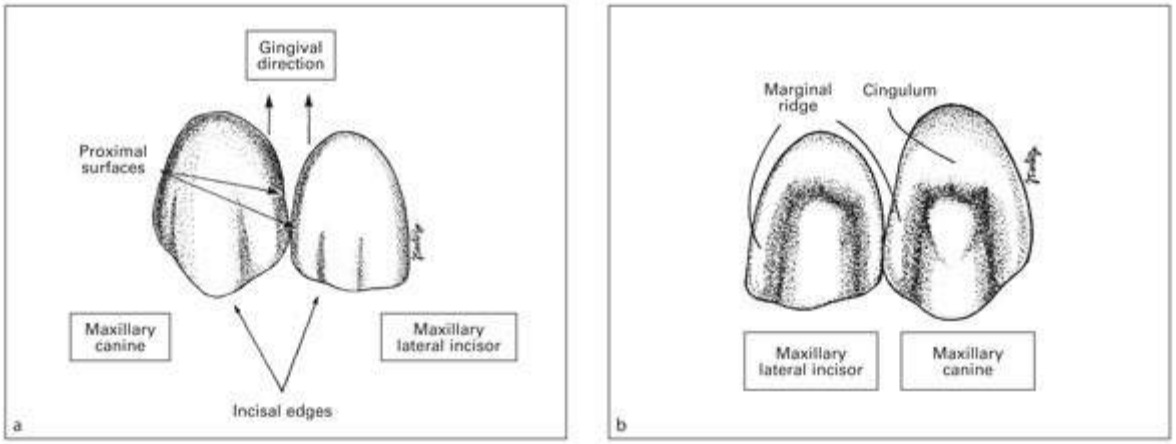


Fig 7-3 Directions, features, and tooth surfaces of anterior teeth. *(a)* Facial view. *(b)* Lingual view.

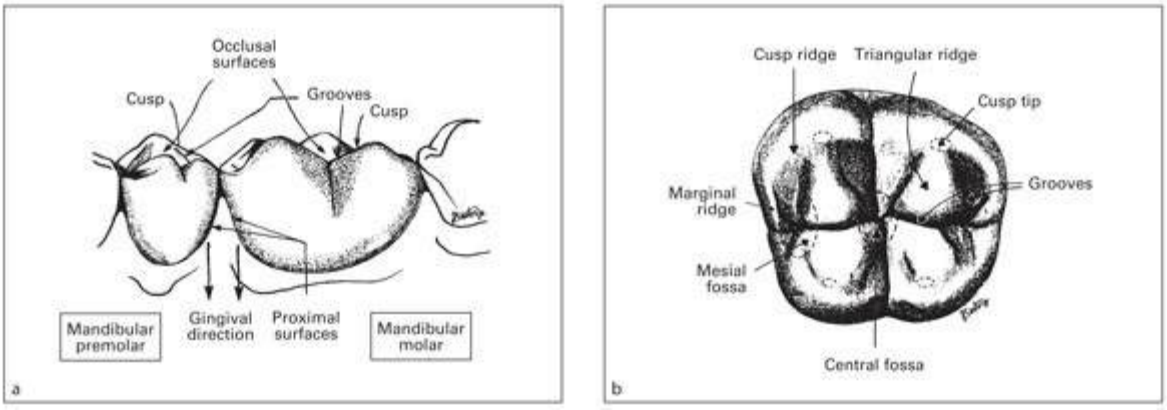


Fig 7-4 Directions, features, and tooth surfaces of posterior teeth. *(a)* Lingual view of mandibular right second premolar and first molar. *(b)* Occlusal view of mandibular right molar.

The anatomical contour of anterior teeth is less complicated than that of posterior teeth, in which the occlusal surfaces are characterized by grooves, cusp tips and

ridges, marginal ridges, and fossae. *Marginal ridges* (both mesial and distal) border the lingual surfaces of anterior teeth (see Fig 7-3b) and the occlusal surfaces of posterior teeth (see Fig 7-4b). A *groove* is a linear channel between enamel elevations, such as cusps and/or ridges. A *fissure* is a developmental linear cleft usually found at the base of a groove; it is commonly the result of the lack of fusion of the enamel of adjoining dental cusps or lobes. A *pit* is a small depression in enamel, usually located in a groove and often at the junction of two or more fissures. A *fossa* is a hollow, rounded, or depressed area in the enamel surface of a tooth. For example, a mesial fossa lies just distal to a mesial marginal ridge[1] (see Fig 7-4b).

Because of the advent of bonding restorative materials to teeth, walls of cavity preparations are less distinct than they are in preparations for restorations that are retained by mechanical undercuts in the preparation walls or by nonadhesive cements. The walls of cavity preparations, however, are generally referred to by the same terms as the surface features of the teeth (eg, the gingival and distal walls) (Figs 7-5 to 7-9). Exceptions are the *pulpal wall* (or floor), which is only in the occlusal portion of a preparation and is the wall adjacent or nearest to the pulp chamber of the tooth (see Figs 7-6 and 7-7), and the *axial wall*, which, in all other areas of a preparation, is the wall adjacent or nearest to the pulp chamber or pulp canal(s) and is approximately parallel to the long axis of the tooth[1] (see Figs 7-5, 7-6, and 7-9).

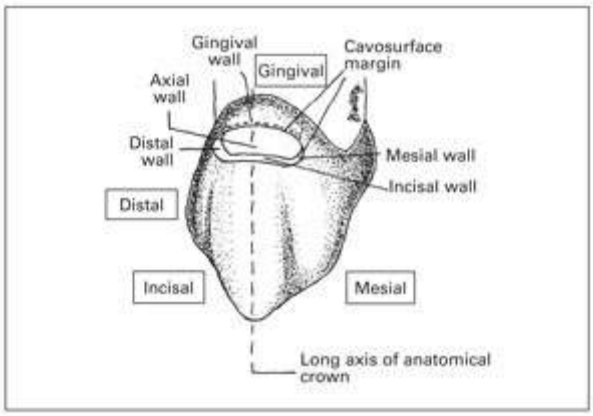


Fig 7-5 Class 5 cavity preparation in an anterior tooth (maxillary canine, facial view). In a posterior tooth, there would be an occlusal wall instead of an incisal wall.

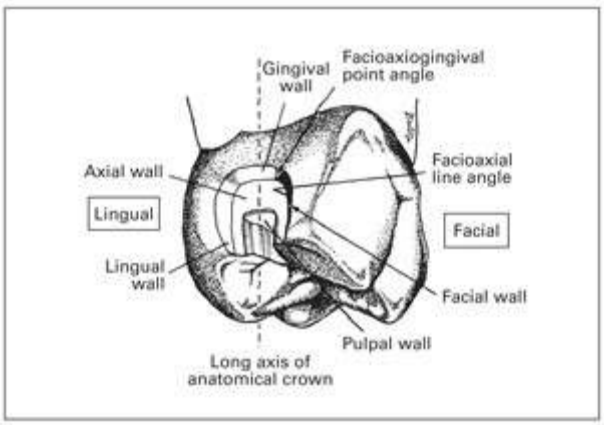


Fig 7-6 Class 2 cavity preparation (maxillary molar, proximal view).

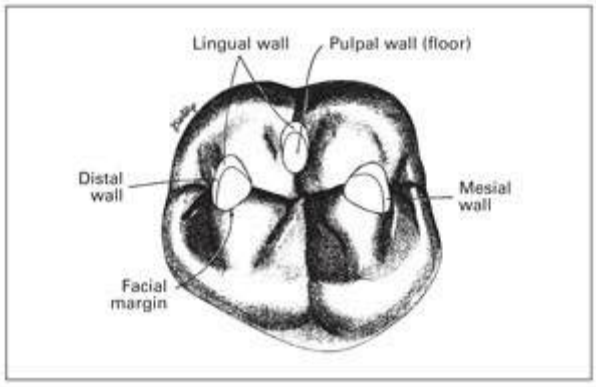


Fig 7-7 Class 1 cavity preparations in a posterior tooth. The occlusal surface of this mandibular right molar is viewed slightly from the facial aspect (occlusofacial), so the facial wall is hidden from view.

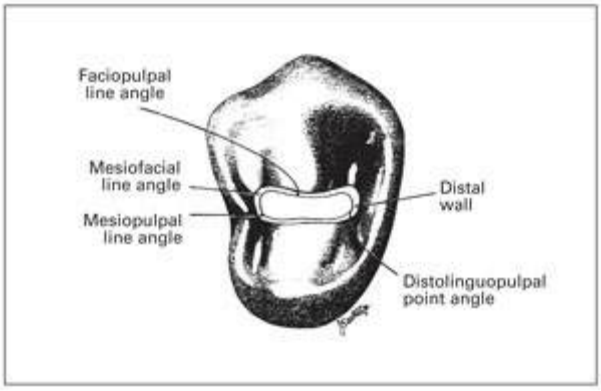


Fig 7-8 Class 1 cavity preparation in a posterior tooth (maxillary premolar, occlusal view). In a direct restoration (such as for amalgam), the facial and lingual walls would be parallel or convergent for retention of the amalgam. The walls of a preparation for bonded resin composite could diverge as shown here or by considerably more, because the restoration would be bonded to the enamel and dentin.

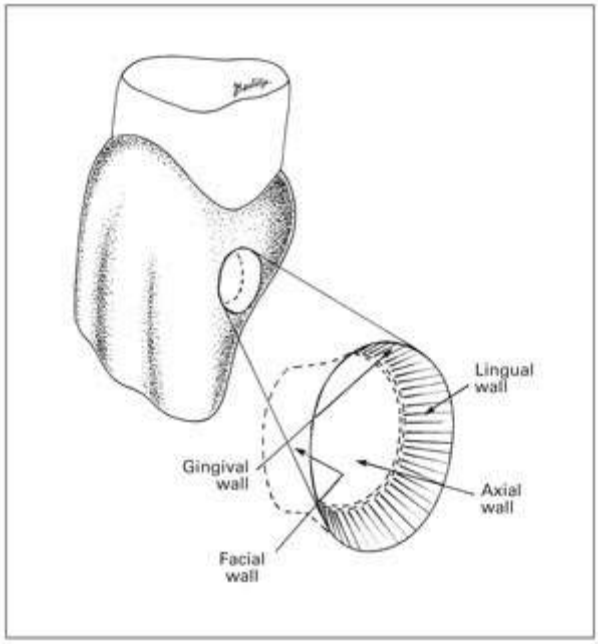


Fig 7-9 Class 3 cavity preparation (maxillary incisor, mesiofacial view). The preparation is for a bonded, tooth-colored restoration.

The junction of two walls in a cavity preparation is called a *line angle*.[1] Again, in preparations for bonded restorations, line angles may not be well defined, but the names for line angles may be used to refer to general areas of the preparation. For example, the meeting of the facial and axial walls forms the facioaxial (or axiofacial) line angle (see Fig 7-6). Similarly, the junction of three walls is referred to as a *point angle*. For example, the junction of the facial, axial, and gingival walls creates the facioaxiogingival (or axiofaciogingival or gingivofacioaxial) point angle. Again, the junction of two walls is often rounded, so it does not actually form a line, but it is still referred to as a line angle; likewise, a point angle is usually not a sharp point.

The *margins* (or *cavosurface angles*1) of a preparation, which are formed by the junction of a cavity wall and an external tooth surface, are identified by the names of the adjacent walls (eg, incisal margin, mesial margin, or gingival margin).

The *anatomical crown* of a tooth is the portion that extends from the cementoenamel junction, or cervical line, to the occlusal surface or incisal edge; it is covered by enamel (Figs 7-10 and 7-11). The *clinical crown* is the portion that is visible in the oral cavity.[1] Depending on the tooth, the clinical crown may include only part of the anatomical crown (see Fig 7-10), or it may include all of the anatomical crown and part of the root (see Fig 7-11).

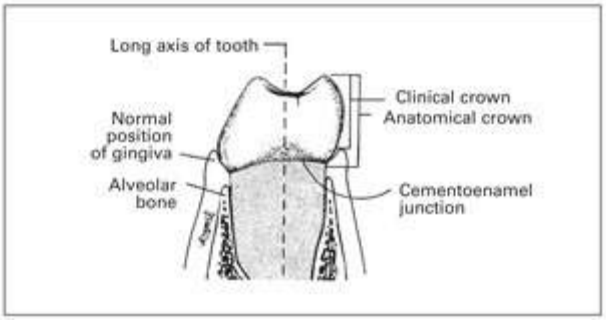


Fig 7-10 Anatomical and clinical crowns of a mandibular molar with the periodontal attachment at the normal, healthy level.

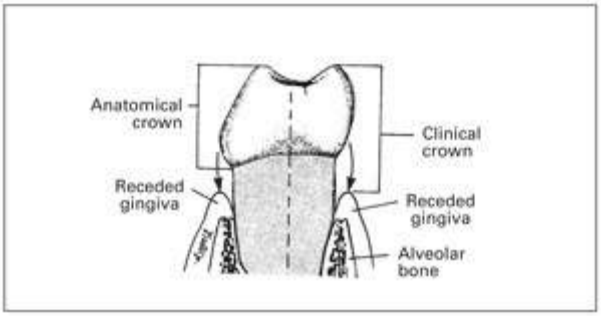


Fig 7-11 Anatomical and clinical crowns of a mandibular molar with the periodontal attachment at a more apical level (periodontal recession). With loss of gingival height comes an increase in the length of the clinical crown, but the anatomical crown, which is defined by the cementoenamel junction (or cervical line), stays constant. The occlusogingival dimension of the anatomical crown can be reduced by loss of occlusal tooth structure from wear or erosion.

Classification of caries lesions and tooth preparations

Near the beginning of the 20th century, G. V. Black,[2] who is known as the father of operative dentistry, classified caries lesions into groups according to their locations in permanent teeth. The same classification is used to refer to cavity preparations, because the location of carious tooth structure is a major factor in the design of the cavity preparation and the selection of instruments (Fig 7-12).

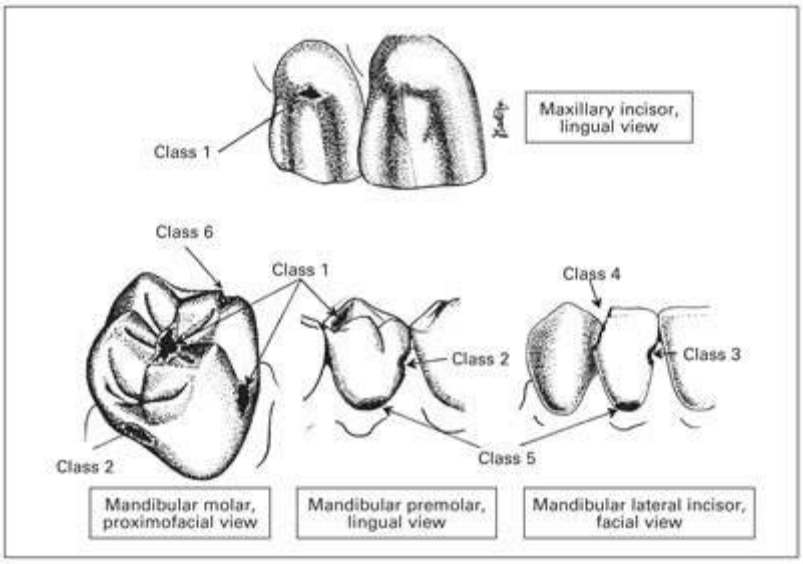


Fig 7-12 Classification of cavity preparations.

Class 1 (I) lesions occur in pits and fissures on the facial, lingual, and occlusal surfaces of molars and premolars and, less often, on the lingual surfaces of maxillary anterior teeth (most frequently lateral incisors, less frequently central incisors, rarely canines) (see Figs 7-7, 7-8, and 7-12).

Class 2 (II) lesions occur in the proximal surfaces of the posterior teeth (molars and premolars). If a proximal surface of a posterior tooth is involved in a restoration, it is a Class 2 restoration (see Figs 7-6 and 7-12).

Class 3 (III) lesions occur in the proximal surfaces of anterior teeth (central and lateral incisors and canines). Class 3 cavities do not involve an incisal angle (see Figs 7-9 and 7-12).

Class 4 (IV) lesions occur in the proximal surfaces of anterior teeth when the incisal angle requires restoration. The angle may have to be removed because of its fragility or for proper placement of the restoration, or it may have been fractured by trauma (Fig 7-13; see also Fig 7-12).

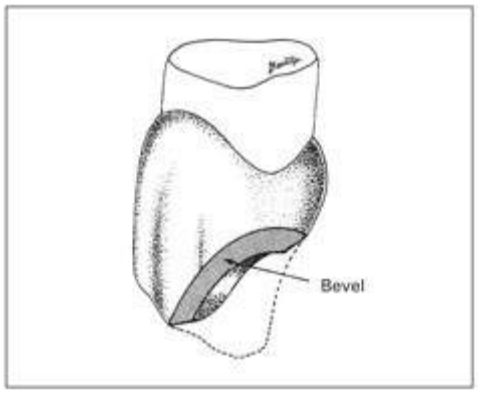


Fig 7-13 Class 4 preparation for a bonded, tooth-colored restoration (maxillary incisor, facial view).

Class 5 (V) lesions occur in smooth facial and lingual surfaces in the gingival third of teeth. Class 5 lesions begin close to the gingiva and may involve a cementum or dentin surface as well as enamel (see Figs 7-5 and 7-12).

Class 6 (VI) lesions are in pits or wear defects on the incisal edges of anterior teeth or the cusp tips of posterior teeth (Fig 7-14; see also Fig 7-12).

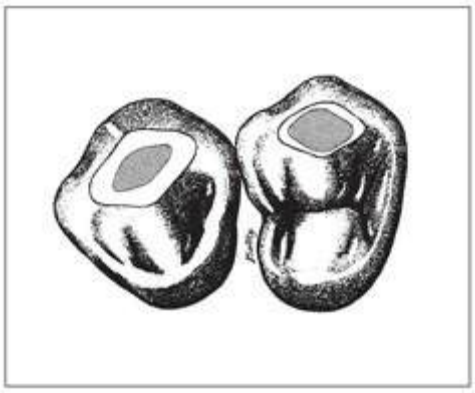


Fig 7-14 Class 6 preparations in the incisal edge of a maxillary canine and the cusp tip of a premolar (incisal/occlusal view). The dotted area of the preparation represents dentin; the clear area of the preparation represents enamel. The preparations have no mechanical, or undercut, retention; they are for bonded, tooth-colored restorations.

In addition to being named for their classifications, cavity preparations and restorations are named for the tooth surfaces involved. For example, a restoration involving the mesial and occlusal surfaces of a posterior tooth is called a *mesio-occlusal Class 2 restoration*; simply saying mesio-occlusal restoration identifies it as a Class 2 restoration because the proximal surface of a posterior tooth is involved. A preparation or restoration involving the mesial, occlusal, distal, and facial surfaces of a posterior tooth is called a *mesio-occlusodistofacial preparation* or *restoration*.

For brevity's sake, the names of the surfaces are often abbreviated (distal, *D*; lingual, *L*; facial, *F*; mesial, *M*; incisal, *I*; occlusal, *O*; buccal, *B*). A mesio-occlusal restoration in a posterior tooth is abbreviated *MO*, and a distolingual restoration in an anterior tooth is abbreviated *DL*.

Black's steps in cavity preparation

Treatment modalities for dental caries other than surgical removal of lesions are discussed in chapter 5. When it has been determined that nonsurgical means of

treating caries lesions will not suffice, however, restorative therapy is indicated. This involves the surgical removal of carious tooth structure and restoration of the tooth to its original anatomical form with a suitable restorative material. The design of the cavity preparation is determined first by the location of the caries lesion(s) in the tooth. The shape or outline of the cavity preparation, as it meets the external surface of the tooth, is referred to as *outline form*. Other factors influencing the design include the need to obtain access for the instruments as the operator is preparing the cavity or placing the restoration (*convenience form*) and the need to provide retention for the restorative material (*retention form*). Also required is resistance to stresses on the restoration and the tooth from forces of biting and chewing (*resistance form*). Because cavity preparation is a surgical procedure in which a mistake can mean injury to living tissue, it is essential that the operator be knowledgeable and highly skilled.

Sequential steps for cavity preparation were established by Black.[2] Black's steps represent a systematic, scientific procedure for efficiency in cavity preparation. Although the technology of bonding restorative materials to enamel and dentin was not available to Black, his steps of cavity preparation are generally as appropriate today as they were when he formulated them:

1. *Establish outline form.* Outline form is based primarily on the location and extent of the caries lesion, tooth fracture, or erosion. In carious teeth, the outline form is established after penetration into carious dentin and removal of the enamel overlying the carious dentin. The extent of carious dentin should be a primary determinant of the outline form of the preparation; the final outline is not established until the carious dentin and, usually, its overlying unsupported enamel have been removed.
2. *Obtain resistance form.* Resistance for the remaining tooth structure and for the restoration must be designed in the preparation so that the restoration will be resistant to displacement and both the tooth and the restoration will be resistant to fracture during function.
3. *Obtain retention form.* Retention may be obtained through mechanical shaping of the preparation to retain the restoration and/or via bonding procedures that attach the restorative material to tooth structure.
4. *Obtain convenience form.* Convenience form allows adequate observation, accessibility, and ease of operation during preparation and restoration of the tooth. Convenience form that involves the removal of sound, strong tooth structure should be limited to that which is necessary.

5. *Remove remaining carious dentin.* Removal of remaining carious dentin applies primarily to that in the deepest part (pulpally) of the preparation. Other carious tooth structure was removed when the outline form was established. Caries removal may necessitate a modification of the outline form.
6. *Finish enamel walls and cavosurface margins.* For indirect restorations (those requiring the making of an impression and fabrication of a stone duplicate of the preparation or the creation of a digital image), finishing involves making the walls relatively smooth. For direct and indirect restorations not utilizing bonding, finishing involves removing any unsupported, weak, or fragile enamel and making the cavosurface margin smooth and continuous to facilitate finishing of restoration margins. For bonded resin composite restorations, enamel that is not supported by dentin and is not going to be exposed to significant occlusal loading is frequently allowed to remain in place and is reinforced by bonding to its internal surface. However, weak, fragile, or unsupported enamel should be removed to prevent fracture due to polymerization shrinkage.
7. *Clean the preparation.* Black referred to this step as "performing the toilet of the cavity." It includes washing or scrubbing away any debris in the preparation and drying the preparation. Afterward, the cavity is inspected for any remaining debris, fragile enamel, and demineralized tooth structure and altered if necessary; then the restoration is placed.

Instrumentation

Hand instruments

Black[2] organized not only the classification of cavity preparations and their parts but also the naming and numbering of hand instruments. *Cutting instruments*, which he also called *excavators*, were to be used in shaping the tooth preparation. All other hand instruments are grouped into the noncutting category.

Metals

For many years, carbon steel was the primary material used in hand instruments for operative dentistry, because carbon steels were harder and maintained sharpness better than stainless steels. Stainless steels, which are more corrosion resistant than carbon steels, are now the preferred materials for hand instruments, because all instruments must be sterilized with steam or dry heat between patients and because

the properties of stainless steels have improved. There are literally hundreds of formulas for stainless steels,[3] all based on iron and incorporating a significant amount of chromium, some carbon, and very often nickel. Chromium imparts corrosion resistance and brightness to the metal; carbon imparts hardness.

Cutting instruments

Before rotating instruments were available, dentists could cut well-shaped cavity preparations using sharp hand instruments alone. The process was slow. The advent of the dental handpiece in 1871,[4] first attached to a foot-operated engine, allowed increased speed of tooth preparation. Most tooth preparation today is accomplished with rotary instruments, but hand cutting instruments are still important for finishing many tooth preparations. Few preparations involving a proximal surface can be completed properly without the use of hand cutting instruments. It is crucial that hand instruments used for cutting tooth structure or carving restorative materials be sharp.

Design. Hand cutting instruments are composed of three parts: handle (or shaft), shank, and blade[2] (Fig 7-15). The *primary cutting edge* of a cutting instrument is at the end of the blade (called the *working end*), but the sides of the blade are usually beveled and also may be used for cutting tooth structure (Fig 7-16). The *shank* joins the blade to the handle of the instrument and is angled to keep the working end of the blade within 2 to 3 mm of the axis of the handle (Fig 7-17). This angulation is intended to provide balance, so that when force is exerted on the instrument it is not as likely to rotate, which would decrease the effectiveness of the blade and could possibly cause damage to the tooth or soft tissue. Figure 7-17a illustrates an instrument that has a single angle at the junction of the blade and the shank. Because the working end of the blade is not aligned with the handle, the instrument is said to be out of balance. Such an instrument may still be useful in tooth preparation. Its blade will usually be relatively short, and it will usually be used with minimal force. Figure 7-17b shows a shank that has two angles to bring the cutting edge into near alignment with the long axis of the handle to provide balance.

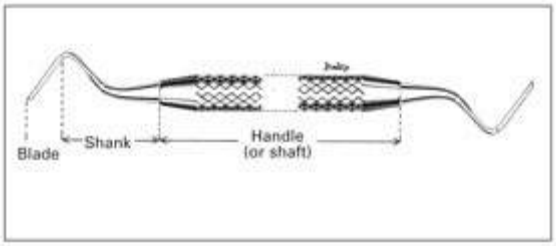


Fig 7-15 Components of a hand instrument. Although the handle is also called a *shaft*, that designation is little used.

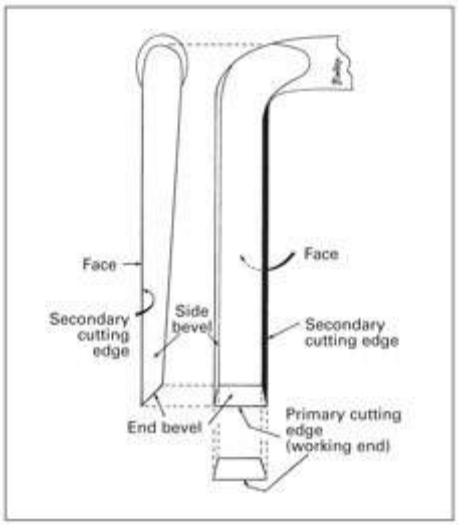


Fig 7-16 Blade bevels. Most hand cutting instruments not only have a bevel on the end of the blade but also have bevels on the sides. Although most of the work of a hand cutting instrument is accomplished with the end of the blade, the sides may also be used to plane or scrape walls and margins. (In the illustration on the right, the blade is lying face down.)

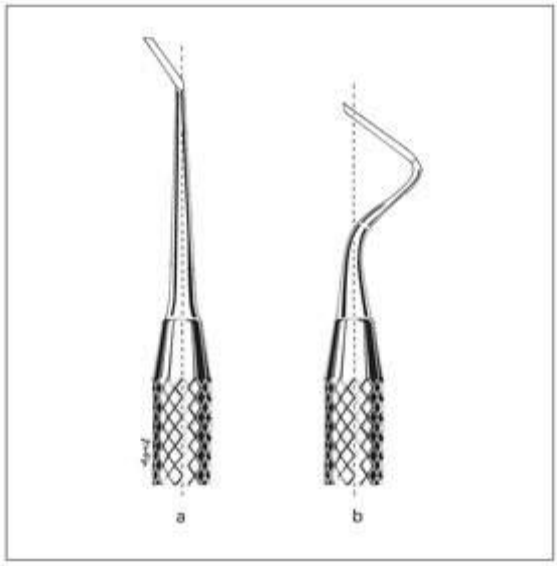


Fig 7-17 The shanks of instruments have multiple angles to keep the working end of the instrument within 2 to 3 mm of the long axis of the handle. *(a)* The working end of this instrument is not close to the long axis of the handle, and the instrument is therefore not balanced. *(b)* The shank of this instrument has two angles in it so that the working end is brought near (within 2 mm) to the long axis of the handle; this provides balance to facilitate control of the instrument during the application of force. The instrument is said to be contra-angled.

A variety of handle configurations are available (Fig 7-18). Padded handles (Fig 7-18b) are said to increase operator comfort and grip during use. Most metal handles today are round and have knurled areas for improved grip. The standard metal handle has a diameter of approximately ¼ inch (6.4 mm). Although little

research support can be found, handles with larger diameters, such as the 3/8-inch (9.5-mm) diameter handle illustrated in Fig 7-18c, are said to be more ergonomic and less likely to contribute to the development of carpal tunnel syndrome. A handle with an intermediate diameter (5/16 inch or 7.9 mm) is also available. The larger diameters are encouraged primarily for dental hygienists, who spend a large part of their day using hand instruments. A drawback to the use of larger handles in operative dentistry is the space they consume in an instrument tray.

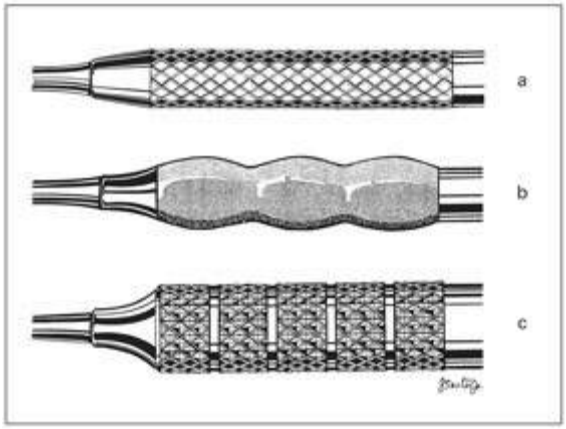


Fig 7-18 Instrument handle configurations. Instrument handles are available from most manufacturers in a variety of designs and diameters. *(a)* Standard stainless steel handle with a diameter of approximately 6.4 mm. *(b)* Padded handles are available; the diameter of the illustrated padded handle is approximately 8 mm. *(c)* Handles with larger diameters are said to be more ergonomic; the diameter of the one illustrated is approximately 9.5 mm.

Nomenclature. The terminology organized by Black[2] in the early part of the last century is still used today, with minor modifications. Most names Black assigned to cutting instruments were based on the appearance of the instrument, such as hatchet, hoe, spoon, and chisel. For an instrument that did not have the appearance of a commonly used item, Black based the name on the intended use (eg, gingival margin trimmer). Black called all cutting instruments used for tooth preparation *excavators*, and he referred to instruments as hatchet excavators, spoon excavators, etc. The term *excavator* is still applicable, but in the day-to-day language of operative dentistry, it is little used. In catalogs of instruments, however, cutting instruments are often indexed as excavators.

Black combined the name of each instrument with a designation of the number of angles in the shank of the instrument. Shanks may be *straight*, *monangle* (one angle), *binangle* (two angles), *triple-angle* (three angles), or *quadrangle* (four angles) (Fig 7-19). The term *contra-angled* refers to a shank in which two or more angles are necessary to bring the working end into near alignment (within 2 to 3 mm) with the axis of the handle (see Fig 7-17b).

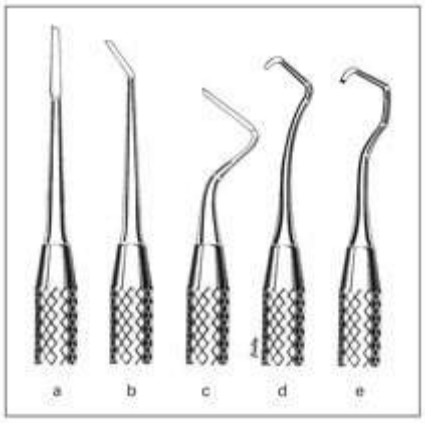


Fig 7-19 Instruments classified by the number of angles in the shank: *(a)* straight; *(b)* monangle; *(c)* binangle; *(d)* triple-angle; *(e)* quadrangle.

Hatchet. In a *hatchet* (also called an *enamel hatchet*), the blade and cutting edge are on a plane with the long axis of the handle; the shank has one or more angles (Figs 7-19e, 7-20a, 7-21, and 7-22). The face (see Fig 7-16) of the blade of the hatchet will be directed either to the left or the right in relation to the handle, and the instrument is usually supplied in a double-ended form. Therefore, there are left-cutting and right-cutting ends of the double-ended hatchet.



Fig 7-20*(a)* Binangle hatchet. *(b)* Binangle spoon. A double-ended hatchet or spoon would have a left-cutting end and a right-cutting end (see Fig 7-22).



Fig 7-21*(a and b)* Monangle hatchets (left-cutting).

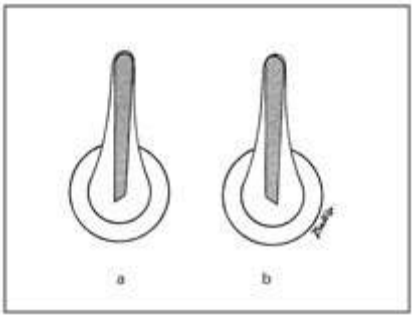


Fig 7-22 End view of binangle hatchets, paired: *(a)* right-cutting; *(b)* left-cutting. A double-ended binangle hatchet has left-cutting and right-cutting ends.

Chisel. A *chisel* has a blade that is either aligned with the handle (Figs 7-19a, 7-23, and 7-24d), slightly angled (Figs 7-19b, 7-24a, and 7-24b), or curved (Fig 7-24c) from the long axis of the handle, with the working end at a right angle to the handle.



Fig 7-23 Straight chisel with bevels on the sides of the blade, to give secondary cutting edges, as well as on the end (primary cutting edge).

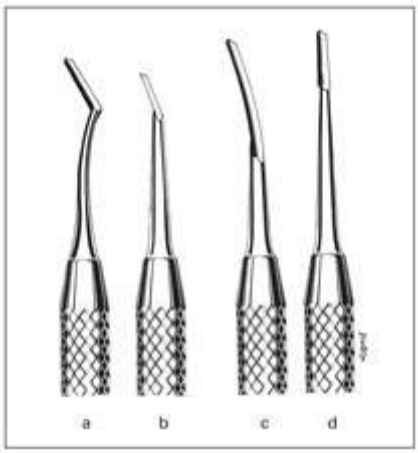


Fig 7-24 Chisels: *(a)* binangle; *(b)* monangle; *(c)* Wedelstaedt; *(d)* straight. The blades for *a*, *c*, and *d* are slightly rotated to visualize the face as well as the side bevel.

Hoe. A *hoe* has a cutting edge that is at a right angle to the handle, like that of a chisel. However, its blade has a greater angle from the long axis of the handle than does the chisel; its shank also has one or more angles (Figs 7-15, 7-19c, and 7-25). A general guideline for distinguishing between a hoe and a chisel will be given later in the chapter.



Fig 7-25 Hoes: *(a)* monangle; *(b)* binangle. The blade of a hoe has an angle from the long axis of the handle of greater than 12.5 centi-grades (45 degrees); in contrast, the blade of a chisel will have an angle from the long axis of the handle of 12.5 centigrades (45 degrees) or less (see Figs 7-33 and 7-34).

Spoon. The blade of a *spoon* is curved, and the cutting edge at the end of the blade is in the form of a semicircle (Figs 7-20b and 7-26b); this gives the instrument an outer convexity and an inner concavity that make it look somewhat like a spoon. Like the hatchet, the spoon has a cutting edge at the end of its blade that is parallel to the handle of the instrument; therefore, there are left-cutting and right-cutting spoons. The shank of some spoons holds a small circular, or disk-shaped, blade at its end,

and the cutting edge extends around the disk except for its junction with the shank; these are called *discoid spoons* (Figs 7-26a and 7-26c).

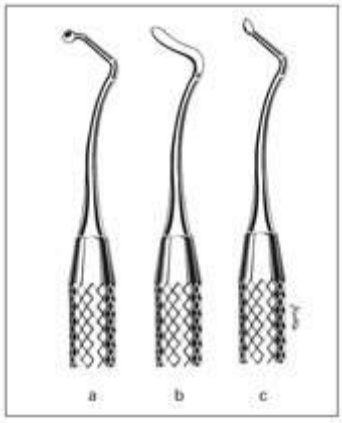


Fig 7-26 Spoons: *(a)* triple-angle discoid spoon; *(b)* binangle spoon (or *regular spoon* or *banana spoon*) ; *(c)* binangle discoid spoon. Spoons are used in tooth preparation for removing (or "spooning out") carious dentin.

Gingival margin trimmer. A *gingival margin trimmer* is similar to an enamel hatchet, except that the blade is curved and the bevel for the cutting edge at the end of the blade is always on the outside of the curve; the face of the instrument is on the inside of the curve (Figs 7-27 and 7-28). Gingival margin trimmers, like hatchets and spoons, come in pairs (left cutting and right cutting) (see Fig 7-27), but there are also mesial gingival margin trimmers and distal gingival margin trimmers (see Fig 7-28). Thus, a set of gingival margin trimmers is composed of four instruments: left-cutting and right-cutting mesial gingival margin trimmers and left-cutting and right-cutting distal gingival margin trimmers. Because these are usually double-ended instruments, one instrument is a mesial gingival margin trimmer (with left- and right-cutting ends), and the other is a distal gingival margin trimmer (with left- and right-cutting ends). Figure 7-28a illustrates a mesial left-cutting gingival margin trimmer. Figure 7-28b illustrates a distal left-cutting gingival margin trimmer. Contrasted with these is a right-cutting hatchet (see Fig 7-28c). Gingival margin trimmers have many uses in addition to trimming gingival margins (Fig 7-29).

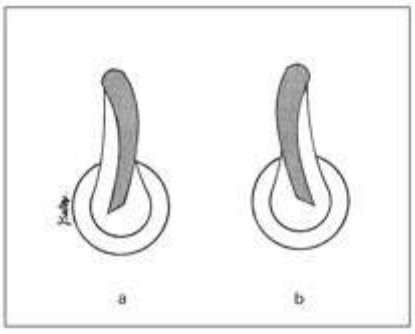


Fig 7-27 End view of gingival margin trimmers, paired: *(a)* right-cutting, *(b)* left-cutting. A double-

ended gingival margin trimmer has both left-cutting and right-cutting ends, but there must be two double-ended gingival margin trimmers to complete a set, one double-ended mesial gingival margin trimmer and one double-ended distal gingival margin trimmer.



Fig 7-28*(a)* Left-cutting mesial gingival margin trimmer. *(b)* Left-cutting distal gingival margin trimmer. *(c)* Right-cutting binangle hatchet.

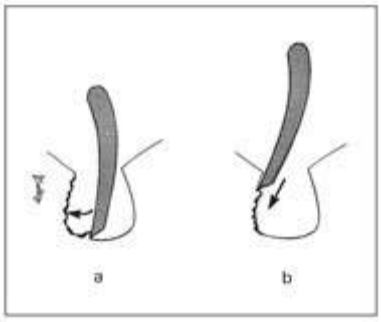


Fig 7-29*(a)* Gingival margin trimmer being used in a proximal box of a Class 2 preparation with a horizontal (left or right) stroke to scrape (plane) a gingival wall and margin. *(b)* Gingival margin trimmer being used with a vertical, or chopping, stroke to plane a facial or lingual wall and margin. A hatchet could be used in a similar way.

Off-angle hatchet. Black's instrument names apply to instruments that have cutting edges that are either parallel or at a right angle to the handle. Instruments have been developed that have blades rotated 45 degrees from the plane of the long axis of the handle; these are called *off-angle hatchets*.

Usage. Hand cutting instruments are, for the most part, made in pairs, and, as with the gingival margin trimmers, most instruments used today are double-ended and will have one of the pair on each end (see Figs 7-15, 7-22, and 7-27). A cutting instrument may be used with horizontal strokes, in which the long axis of the blade is directed at between 45 and 90 degrees to the surface being planed or scraped (see Fig 7-29a), or with vertical or chopping strokes, in which the blade is nearly

parallel to the wall or margin being planed (see Fig 7-29b). For horizontal (scraping) or vertical (chopping) strokes, the acute angle of the cutting edge is intended for use. The acute angle is the junction of the face of the blade with the bevel; in other words, the bevel is on the back of the blade, not the face of the blade. A double-ended hatchet, gingival margin trimmer, or spoon will have one end that is designated as right cutting and one that is designated as left cutting. In a double-ended hoe, in addition to allowing vertical or chopping strokes, one end is intended for pulling strokes (beveled end or end with distal bevel) and the other is intended for pushing strokes (contrabeveled end or end with mesial bevel).

The cutting edges of most hand cutting instruments used today are single beveled, as are all of those described here (see Fig 7-16). Double-beveled, or bi-beveled, cutting edges are also available but have limited application in contemporary operative dentistry (Fig 7-30). These instruments usually have narrow blades and are used for tasks such as adding mechanical retention points in areas of preparations that cannot be reached by a bur.

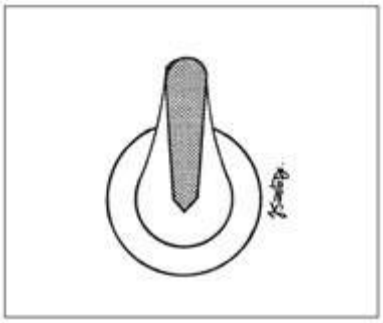


Fig 7-30 Bi-beveled cutting edge. These instruments are useful in placing retention points in some direct gold (gold foil) preparations but have little use in any of the preparations described in this book.

Numeric formulas. The configuration of the shanks combined with the appearance of the blade or the use of the instrument produces names such as straight chisel, monangle chisel, binangle hoe, and triple-angle hatchet. These are descriptive terms, but they are imprecise because they do not indicate sizes or angles. For more complete identification of hand cutting instruments, Black[2] developed a system of assigning numeric formulas to instruments (Figs 7-31 and 7-32). The formulas make use of the metric system. For designating the degree of angulation, *centigrades* are used. Centigrades are based on a circle divided into 100 units (Fig 7-33), as opposed to the 360-degree circle ordinarily used to designate angles. In a centigrade circle, a right angle has 25.0 centigrades.

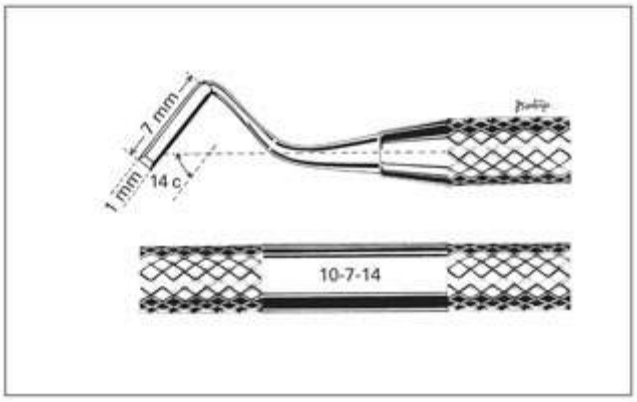


Fig 7-31 Black's three-number formula for instruments that have a primary cutting edge (working end) that is at a right angle (90 degrees) to the long axis of the blade: The first number is the width of the blade in tenths of a millimeter; the second number is the length of the blade in millimeters; and the third number is the blade angle, the angle the blade makes with the long axis of the handle, in centigrades (c). The complete name of the instrument illustrated would be binangle hatchet, 10-7-14. The formula would be the same if the blade were rotated 90 degrees on the shank to form a hoe, but the name would be different. Assuming the instrument illustrated is double-ended, the right-cutting end is shown.

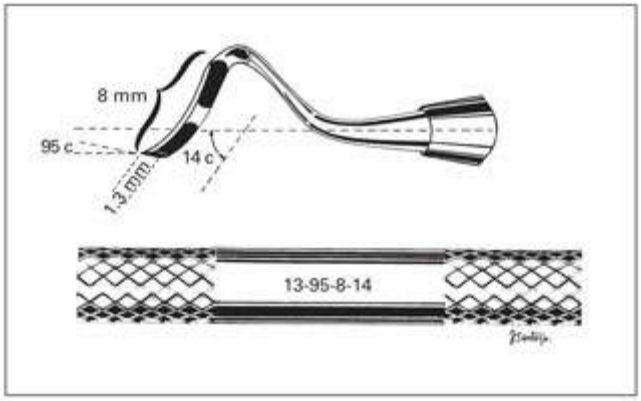


Fig 7-32 Black's four-number formula for instruments that have a primary cutting edge (working end) that is not at a right angle to the long axis of the blade: The first number is the width of the blade in tenths of a millimeter; the second number is the cutting edge angle, the angle the primary cutting edge makes with the long axis of the handle, in centigrades (c); the third number is the length of the blade in millimeters; and the fourth number is the blade angle, the angle the blade makes with the long axis of the handle, in centigrades. Illustrated is the right-cutting end of a distal gingival margin trimmer, 13-95-8-14.

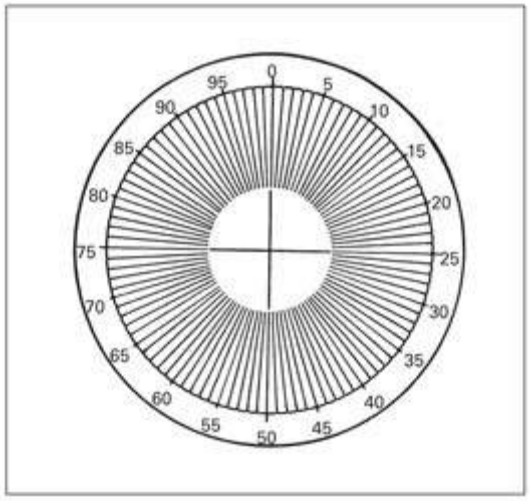


Fig 7-33 Centigrade scale. The circle is divided into 100 units.

Three-number formula. For instruments in which the primary cutting edge (at the end of the blade) is at a right angle to the long axis of the blade, Black developed a formula that has three numbers (see Fig 7-31). The first number is the width of the blade in tenths of a millimeter; the second is the length of the blade in millimeters, and the third is the angle (in centigrades) made by the long axis of the blade and the long axis of the handle (Fig 7-34).



Fig 7-34 Centigrade scale inset to show angulation indicator of 16.0 centigrades for the blade angle of this hoe (three-number formula). The vertical axis (0.0 centigrades) is the axis of the instrument's handle. If the blade of the instrument were ¼ mm wide and 10.0 mm long, the

formula for the instrument would be 14-10-16.

Four-number formula. For instruments in which the cutting edge at the end of the blade is not at a right angle to the long axis of the blade, such as gingival margin trimmers, Black designed a four-number formula (see Fig 7-32). The first number is the width of the blade in tenths of a millimeter; the second number is the angle (in centigrades) that the primary cutting edge (working end) makes with the axis of the handle (Fig 7-35); the third number is the length of the blade in millimeters; and the fourth number is the angle (in centigrades) that the long axis of the blade makes with the handle. In margin trimmers, a cutting edge angle of greater than 90 centigrades is intended for distal gingival margins (see Fig 7-32); an angle of 85 centigrades or less is intended for mesial gingival margins.

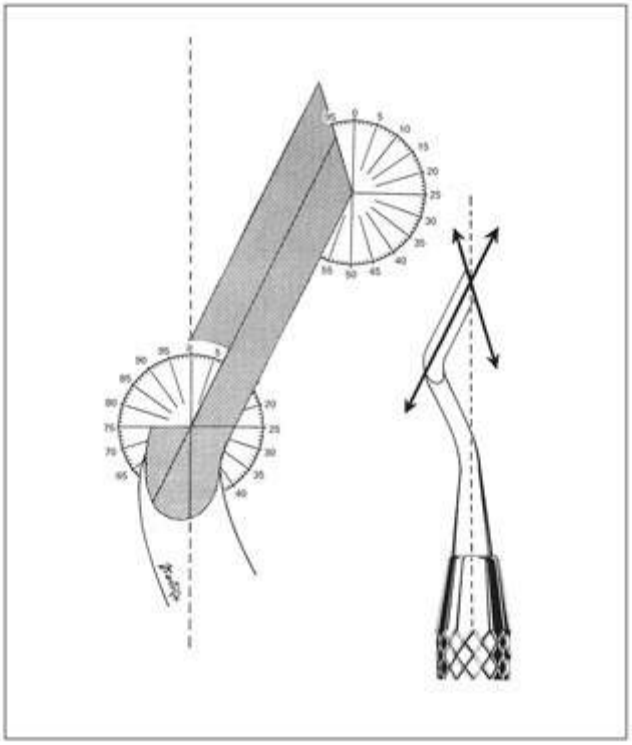


Fig 7-35 Centigrade scales inset to show angulation indicators of 7.0 centi-grades for the blade angle and 95.0 centigrades for the cutting edge angle of this gingival margin trimmer (four-number formula). The vertical axis (0.0 centi-grades) is the axis of the instrument's handle. If the blade were 1.5 mm wide and 10.0 mm long, the formula for the instrument would be 15-95-10-7.

Chisel versus hoe. Although Black defined a chisel as having a blade that is aligned with the handle or slightly curved from it, terminology has evolved so that a chisel may also have a blade that is angled from the handle up to 12.5 centigrades.[5] A chisel with a blade angled more than 3.0 or 4.0 centigrades from the axis of the handle must be binangle for the instrument to be balanced.

If the blade is angled more than 12.5 centigrades, the instrument is defined as a hoe. In a curved or angled chisel or a hoe, a blade with its primary cutting edge (and its face) on the side of the blade toward the handle is said to be *beveled*, or to have a *distal bevel* (see Figs 7-24a to 7-24c); a blade with its primary cutting edge (and its face) on the side of the blade away from the handle is said to be *contrabeveled*, or to have a *mesial bevel* (see Fig 7-19c).

Recommended instrument kit. Black recommended a long set of 96 cutting instruments, a university set of 44 cutting instruments, or a short set of 25 cutting instruments. Because bonding technology and high-speed handpieces were not available, the dental materials of the time were more limited, and a primary restorative material was direct gold; the longevity of restorations depended on the retention and resistance form developed with hand cutting instruments.

With access to advanced materials and technology, current use of hand cutting instruments is greatly diminished. The kit recommended in this chapter (Box 7-1) has only 12 hand cutting instruments (six double-ended instruments). Because it is now recognized that there is no need to plane walls and floors of cavity preparations to smoothness with hand instruments for a restoration to perform well, hand cutting instruments play only a small, albeit important, part in cavity preparation. If burs alone were used for shaping proximal preparations, excessive sound tooth structure would have to be removed from the tooth being restored or the bur would damage the adjacent tooth. Hand cutting instruments enable the dentist to shape and refine small proximal boxes without damaging adjacent teeth.

Box 7-1	**Suggested operative dentistry instrument kit**
A compact assembly of hand instruments that will satisfy most operators' needs during any amalgam, resin composite, glass-ionomer, ceramic, or cast gold restorative procedure is presented here. This kit is especially useful for dental schools and large group practices. Dental students, residents, and practitioners have used the kit, and, although another instrument may have to be added for a specific situation from time to time, the kit will more than suffice for most procedures. The kit was designed with the sequence of most operative procedures in mind.	Therefore, instrument sequence in the kit proceeds from the mirror and explorer for examination, to the plastic instrument used to facilitate dam placement as well as for placement of materials, to tooth preparation instruments, to restoration placement instruments. The kit uses a 26-slot tray with a small well (open, boxlike section) from American Eagle. American Eagle has this tray and others available with customizable color-coded tabs to facilitate replacement of similarly color-coded instruments into the correct positions in the tray.
In slots (in this order, from left to right, with the open well	

to the rear):

- Mirror, no. 5 with handle
- Explorer-periodontal probe, XP23/QOW
- Cotton forceps (college, with serrations)
- Plastic instrument, no. 1-2
- Spoon, discoid, 11½-7-14
- Hatchet, 10-7-14
- Hoe, 12-10-16
- Gingival margin trimmer, 10-80-7-14
- Gingival margin trimmer, 10-95-7-14
- Wedelstaedt chisel, 10-15-3
- Applicator/spatula (American Eagle or Miltex)
- Condenser, SA1 (American Eagle or Miltex)
- Condenser, SA2 (American Eagle or Miltex)
- Condenser, SA3 (American Eagle or Miltex)
- Burnisher, beavertail-ovoid, 2/30
- Burnisher, PKT3
- Barghi no. 1 (paddle-shaped for composite) (American Eagle)
- Carver, cleoid-discoid, UWD5
- Carver, Walls no. 3
- Carver, Hollenback no. ½
- Carver, interproximal (IPC)
- Carver, no. 14L
- Articulating paper forceps
- Carrier, amalgam, medium/large

In well of tray:

- Scalpel handle, no. 3, flat
- Sharpening stone, flat, Arkansas or ceramic
- Tofflemire retainer, straight
- Tofflemire retainer, contra-angle
- Amalgam well, stainless steel, small (American Eagle or Miltex)

Clipped to lid of tray:

- Hemostat, mosquito, 5-inch curved
- Scissors, Quimby

Sterilized separately and available for each operative procedure:

- Anesthetic syringe
- Rubber dam kit (forceps; punch; frame; 1 each of clamps W2A, 27, and 212SA; and 2 W8ASA clamps [Hu-Friedy])
- Brasseler bur block (no. A600) with burs arranged in the following order:
  - Friction-grip burs, no. ⅛, ¼, 1, 2, 33 ½, 56, 169L,170, 329, 330, 7404, OS1F, 7803, 7901, ET9F
  - Latch burs, no. 2, 4, 6, 8
  - Mandrel for pop-on disks

Sterilized separately and available for occasional use:

- Condenser, SA4
- Hemostat, mosquito, 5-inch straight
- Mirror, no. 2 (on handle)
- Proximal contact disks (Thierman Products or Centrex) (see chapter 8)
- Rubber dam clamps, 00, W1A, W8A
- Scaler, McCalls, SM13s-14s
- Spatula, no. 24 (or 324)

Hatchets, hoes, chisels, and gingival margin trimmers have straight cutting edges and are designed to plane enamel and dentinal walls and margins in shaping cavity preparations, especially in areas of the preparation that cannot be reached with a bur. Spoons, on the other hand, have rounded cutting edges; their intended use is the removal of carious dentin. Although slowly rotating round burs are most useful in removing carious dentin, a spoon gives more tactile sensation and is preferred by many operators for evaluating the hardness and penetrability of dentin in tooth

preparation.

Noncutting instruments Non–tooth-cutting hand instruments are similar in appearance to cutting instruments, except that the blade used for tooth preparation is replaced with a part that has a totally different use. In noncutting instruments such as burnishers and condensers, the blade is replaced by a *nib* or *point*.[2] The flat end of the nib of a condenser is called the *face*. Amalgam carvers have carving blades instead of tooth-cutting blades.

Condensers, carvers, and burnishers are used to place dental amalgam and, to a certain extent, resin composite restorative materials. Plastic filling instruments are used to place resin composite and glass-ionomer materials, provisional restorative materials, and sometimes cavity-basing materials into tooth preparations. Spatulas may be necessary for mixing cavity-lining and cavity-basing materials, provisional restorative materials, and cements for luting inlays, onlays, and crowns.

Amalgam carriers. For dental amalgam restorations, amalgam is placed into the preparation with an *amalgam carrier*, an instrument with a hollow cylinder that is filled with amalgam (Fig 7-36). A plunger operated with a finger lever pushes the amalgam out of the carrier into the preparation.

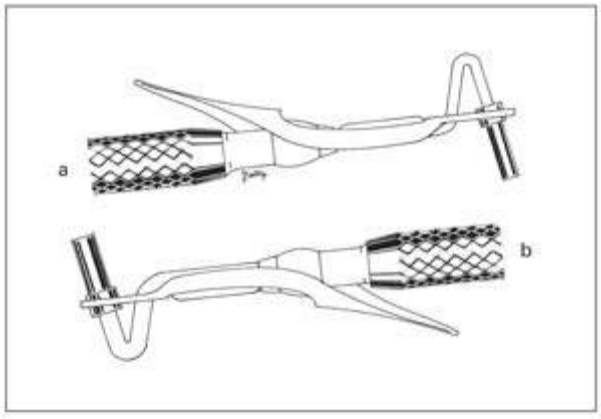


Fig 7-36 Amalgam carriers: *(a)* regular; *(b)* large. Amalgam carriers are usually supplied as double-ended instruments. They are available in several different diameters; for example, mini is 1.5 mm; regular (medium) is 2.0 mm; large is 2.5 mm; and jumbo is 3.0 to 3.5 mm. These are the approximate inside diameters of the cylinders of amalgam carriers and may vary slightly from manufacturer to manufacturer.

Condensers. *Condensers* are used to compress amalgam or to push resin composite or glass-ionomer materials into all areas of the preparation. The working ends, or nibs, of condensers may be any shape, but usually they are round with flat ends (faces). Figure 7-37 shows four round condensers of different sizes and

configurations. Other commonly used condenser nibs are triangular, rectangular, or diamond shaped. Amalgam is condensed by pushing the condenser directly into the preparation and confining the amalgam between the condenser face and the preparation floor through vertical pressure (vertical condensation). The amalgam is condensed against the vertical walls of the preparation (lateral condensation) by angling the nib and using the end for condensation or by lateral, or side-to-side, movements of the condenser, using the sides of the nib to condense the amalgam.

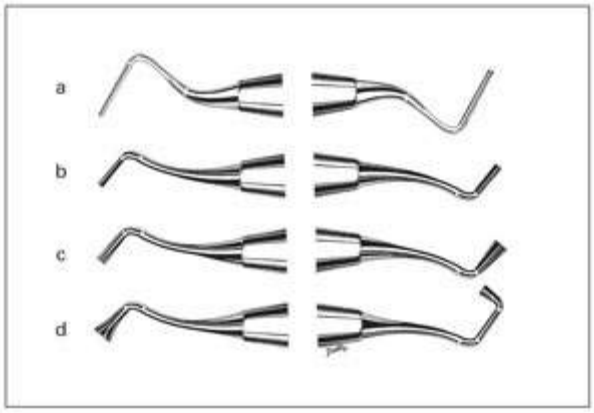


Fig 7-37 Condensers with round faces: *(a)* SA1, with 0.5- and 0.7-mm-diameter faces; *(b)* SA2, with 0.7- and 1.0-mm faces; *(c)* SA3, with 1.5- and 2.0-mm faces; *(d)* SA4, with a 2.5-mm face on the binangle end and a 1.5-mm face on the triple-angle or back-action end.

The condensation pressure applied to the amalgam with a condenser depends on the size of the face and the amount of force used by the operator. For small condensers, such as the SA1 condenser (see Fig 7-37a), little force is needed. The nibs of the SA1 condenser are 0.5 and 0.6 mm in diameter. For larger condensers, such as the SA3 (see Fig 7-37c), with nib diameters of 1.5 and 2.0 mm, a significant amount of force (6 to 8 lbs) gives optimum condensation.

When condensers are used in placing resin composite or glass-ionomer materials, the resin material is not actually condensed but is pushed or patted into all areas of the preparation with the largest condenser face that will fit into the area. Firm pressure will push the condenser into the increment of composite and risks forming a void in the restoration.

Carvers. *Carvers* are used to shape amalgam and resin composite and other tooth-colored materials after they have been placed in tooth preparations. Figure 7-38 shows the shapes of the blades of a cleoid-discoid carver. Figure 7-39 illustrates six commonly used carvers. In general, when a convex amalgam contour is being carved, a concave-shaped carver facilitates the shaping or carving. Likewise, a convex carver facilitates carving of a concave shape. A convex carver may be used

to carve a convex surface; the surface is carved tangentially, with multiple strokes. Whether a carver is used to carve amalgam or resin composite, it is important that the blade be sharp.

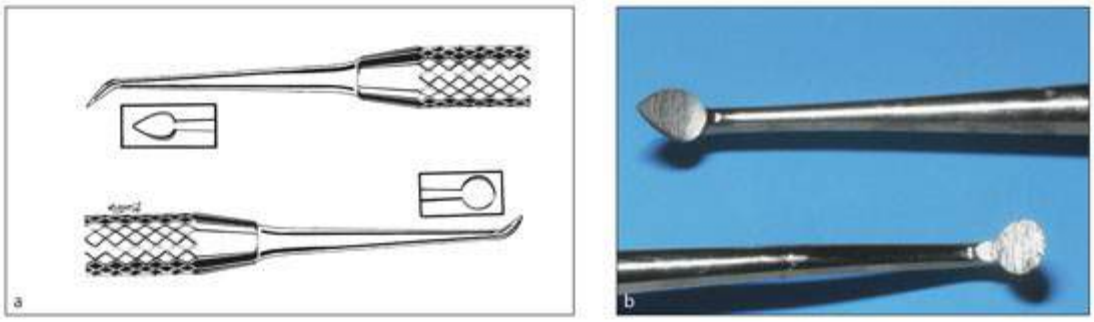


Fig 7-38*(a)* Cleoid-discoid carver: *(top)* cleoid end; *(bottom)* discoid end. This type of carver is a double-ended instrument. *Cleoid* means claw shaped. Both shapes are useful in carving occlusal surfaces of amalgam restorations. The point of the cleoid carver is used to carve the bases of grooves in the occlusal amalgam, and the tip is usually very slightly rounded so that the grooves it carves will not be sharp. *(b)* Cleoid *(top)* and discoid *(bottom)* ends of the cleoid-discoid carver.

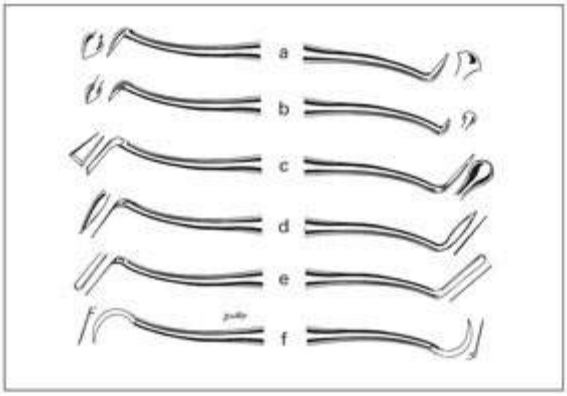


Fig 7-39 Amalgam carvers: *(a)* large cleoid-discoid (Tanner no. 5 [5T]) carver; *(b)* small cleoid-discoid (UWD5) carver; *(c)* Walls no. 3 carver; *(d)* Hollenback no. ½ carver; *(e)* interproximal carver (IPC); *(f)* no. 14L sickle-shaped carver.

The cleoid-discoid (or discoid-cleoid) carvers shown in Figs 7-39a and 7-39b are used primarily for occlusal carving in amalgam restorations. The Walls no. 3 carver (see Fig 7-39c) is useful for carving occlusal surfaces; the end that is shaped like a hoe is also useful for shaping cusps and for carving facial and lingual surfaces of large amalgam restorations. The Hollenback no. ½ carver (see Fig 7-39d) is useful for occlusal, proximal, and axial (facial and lingual) surfaces; several larger Hollenback carvers, with the same general shape, are also available. The interproximal carver (IPC) (see Fig 7-39e) has very thin blades and is extremely valuable for carving proximal amalgam surfaces near the interproximal contact area, as well as those surfaces mentioned for the Hollenback carver (occlusal, proximal,

axial). Other uses for this instrument include placing and shaping resin composite and glass ionomer and pushing retraction cord into a gingival sulcus. The no. 14L carver (see Fig 7-39f) can be used to carve proximal surfaces, or it may be used for carving convex facial and lingual surfaces of very large amalgam restorations. The no. 14L carver has a very strong, hollow-ground triangular blade, so it can be used to remove amalgam overhangs from completely set amalgam.

Although most of the shaping of resin composite restorations should be completed before the material is polymerized and most operators prefer to use rotary instruments for post-polymerization shaping, several amalgam carvers are also useful for carving resin composite. The discoid carvers are especially useful for lingual concavities of anterior teeth; cleoid and discoid carvers and the hoe-shaped end of the Walls no. 3 carver are useful for occlusal surfaces of posterior resin composite restorations. Another carver very useful for resin composite restorations is a disposable scalpel blade (no. 12 or no. 12B blade) mounted in a scalpel handle.

Burnishers. *Burnishers* are used for several functions. The word *burnish* is defined as "to make shiny or lustrous, especially by rubbing; to polish" and "to rub (a material) with a tool for compacting or smoothing or for turning an edge."[6] Burnishing is probably used in all of these ways in dentistry. Two frequently used double-ended burnishers are illustrated in Fig 7-40.

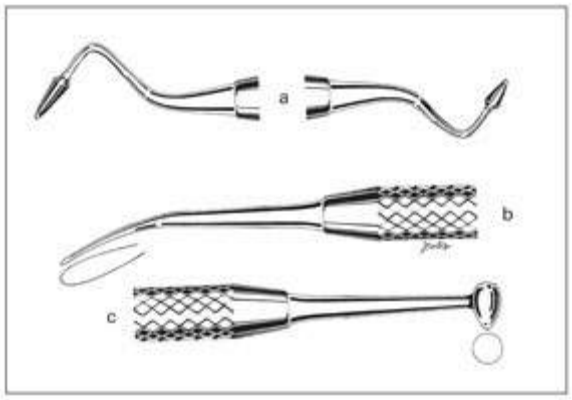


Fig 7-40 Burnishers: *(a)* PKT3 (rounded cone-shaped) burnisher, designed by Peter K. Thomas as a waxing instrument but useful in placing direct restorations as well; its rounded end and cone shape allow it to serve most functions that a small ball-shaped burnisher would serve, plus others; *(b)* beavertail (no. 2) burnisher; *(c)* football or ovoid (no. 30) burnisher. The ovoid burnisher, available in various sizes (eg, 28, 29, and 31), can be used for final condensation of amalgam and the initial shaping of the occlusal anatomy in amalgam. The beavertail and ovoid burnishers are useful for burnishing margins of cast gold restorations.

One use of burnishers is to shape metal matrix bands so that they impart more desirable contours to restorations. Large burnishers are used with considerable

force to pinch off freshly condensed amalgam at the margins, or, in other words, to impart some condensation and to begin shaping the occlusal surfaces of amalgam restorations. After the amalgam has been carved, a burnisher may be used with a gentle rubbing motion to smooth the surface. The PKT3 (P. K. Thomas no. 3) burnisher (see Fig 7-40a) and some other burnishers are also useful for sculpting occlusal anatomy in posterior resin composite restorations prior to polymerization of the resin.

Burnishers are used to "bend" cast gold near the margin to narrow the gap between the gold and the tooth. This closing of a marginal gap is best accomplished with a narrow burnisher, such as the side of a beavertail burnisher, used with heavy force in strokes parallel to the margin but about 1.0 to 1.5 mm away from it. If burnishing is accomplished directly on a thin gold margin, the gold can be bent severely and may break.

Plastic instruments. *Plastic instruments* (or plastic filling instruments) are so named because they were originally designed to use with plastic restorative materials, such as acrylic resins, or other nonmetal restoratives, such as the silicates, used in the middle of the 20th century. The name does not refer to the material from which the instrument itself is constructed. They are currently used to carry and shape tooth-colored restorative materials, such as resin composites and glass-ionomer restorative materials. Many specially designed instruments are available, in a myriad of shapes and sizes, for contouring resin composite and resin-modified glass ionomer prior to curing. A plastic instrument with a large, slightly curved, paddle-shaped blade (eg, Almore Gold Microfil Instrument, Almore; Fig 7-41a) is very useful for placing and contouring large, anterior resin composite restorations or veneers.

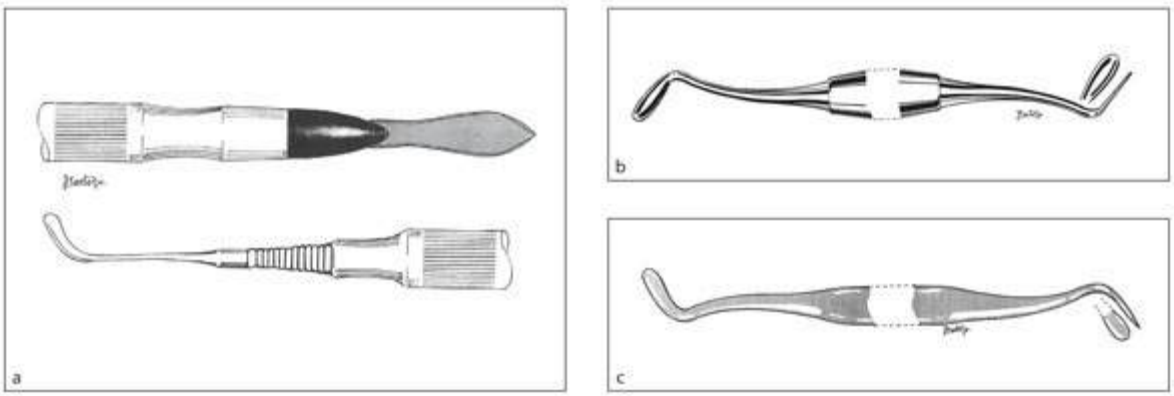


Fig 7-41 *(a)* The Almore Gold Microfil instrument is very useful for placing and contouring large, anterior resin composite restorations or veneers. *(b)* The no. 1-2 plastic instrument, made of stainless steel, is useful for placing a rubber dam, placing and shaping resin composite and other tooth-colored restorative materials, and packing gingival retraction cord into the sulcus around a

crown or abutment preparation before an impression is made. Some cord-packing instruments are similar to the no. 1-2 plastic instrument but may have serrated ends to provide better control of the cord. *(c)* A plastic instrument made of hard plastic, rather than metal, is preferred by some operators for placing resin composites.

A commonly used plastic instrument is the no. 1-2 (Fig 7-41b). The double-ended instrument has a nib or blade on each end, one at a 90-degree angle to the other. Other double-ended plastic instruments have a blade-type nib on one end and a condenser nib on the other. The bladed plastic instruments have many uses in operative dentistry in addition to carrying and contouring restorative materials. The IPC (see Fig 7-39e), for instance, is preferred by some operators for packing knitted cord and placing and shaping resin composite.

These instruments are now available in both hard plastic and metal, and metal instruments are available with several different coatings on blades or nibs to prevent resin materials from sticking to them. The original rationale for using an instrument made of plastic (Fig 7-41c) was to eliminate abrasion of metal by the quartz in resin composites, which caused grayness in the tooth-colored material. Because of changes in the inorganic fillers used in many of today's resin composites, the problem of metal abrasion and graying is unusual and material-specific, so even a stainless steel instrument functions well to carry and shape resin composite.

Cement spatulas. A variety of materials in operative dentistry require mixing, some on a glass slab, others on a paper pad. Several spatulas are available, and they vary in size and thickness (Fig 7-42). The larger cement spatulas were originally designed for mixing luting cements and the smaller spatulas for cavity liners, but since the advent of resin luting cements, the smaller spatulas are frequently used for mixing small amounts of those materials. The thinner spatulas are flexible; the thicker ones are rigid. Selection of a rigid or flexible cement spatula is dependent on the desired viscosity of the cement and personal preference.

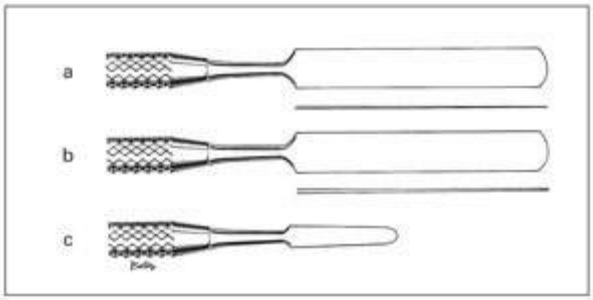


Fig 7-42 Spatulas: *(a)* no. 24, a flexible spatula, is used for luting cements such as zinc phosphate and glass ionomer; *(b)* no. 24A is thicker, for more rigidity; *(c)* no. 313 is used for cavity liners, such as calcium hydroxide liners.

Sharpening of hand instruments

To assess sharpness, the user of the instrument should look at the cutting edge in bright light; the presence of a glint indicates that the edge is dull or rounded (Fig 7-43). Alternatively, the dentist can pull the instrument across hard plastic, such as the handle of a plastic mouth mirror or an evacuator tip. A dull blade will slide across the plastic; a sharp blade will cut into the surface, stopping movement. A specially made, sterilizable, sharpness-testing stick is also available (Fig 7-44) (Dalron Test Stick, American Eagle or Miltex).

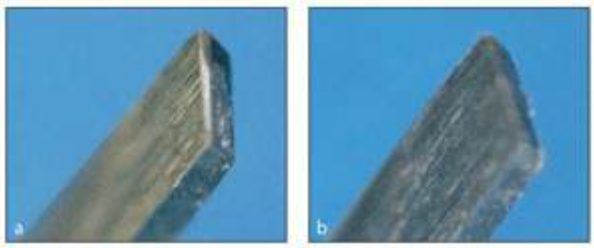


Fig 7-43*(a)* The glint from the cutting edge of this hoe indicates that the blade is quite dull. *(b)* After sharpening, no glint is noticeable.

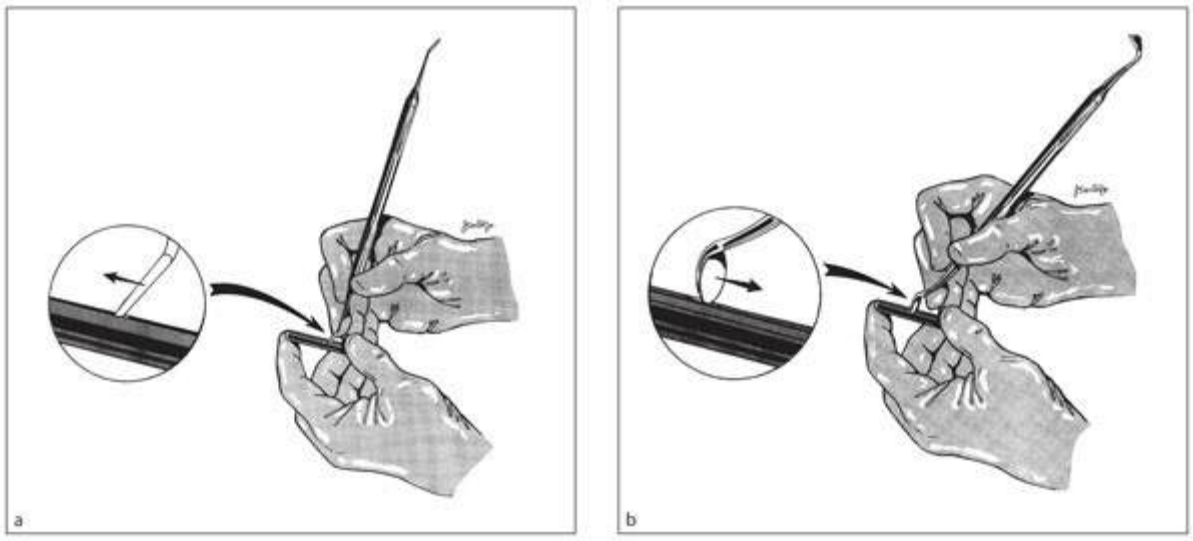


Fig 7-44 The sharpness-testing stick is a hard plastic stick used for testing the sharpness of instruments. To test sharpness, the blade should be applied to the stick at an angle that is similar to that applied during use and pulled or pushed in a direction that is similar to the direction of its intended use. *(a)* Testing the sharpness of a monangle chisel. *(b)* Testing the sharpness of the discoid end of a Walls no. 3 carver.

Sharpening is performed in different ways for different hand instruments. When chisels, hatchets, hoes, and margin trimmers are sharpened, the cutting-edge bevel is placed flat against a flat stone on a stable surface, and the instrument is pushed or pulled so that the acute cutting angle is moved forward, with fairly heavy force on

the forward stroke and with little or no force on the back stroke (Figs 7-45 and 7-46). Usually, unless the instrument has been badly neglected, only two or three forward strokes are required. Because the bevels of these instruments should usually make a 45-degree angle with the face of the blade, the blade should make a 45-degree angle with the surface of the sharpening stone (Figs 7-46 and 7-47).

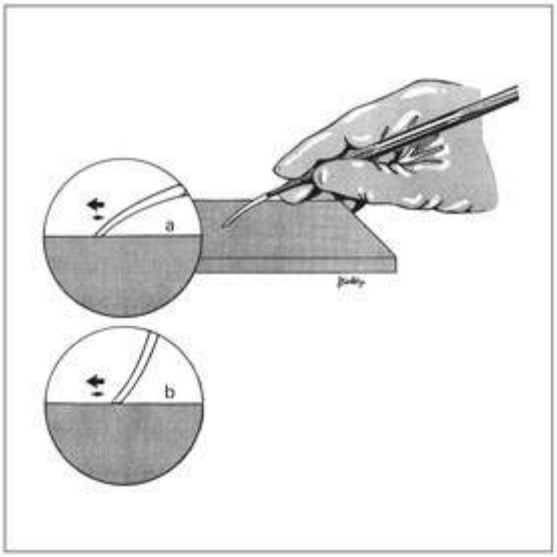


Fig 7-45 Sharpening the two ends of a double-ended Wedelstaedt chisel: *(a)* sharpening the contrabevel end (inside or mesial bevel); *(b)* sharpening the bevel end (outside or distal bevel). The end bevel (for the primary cutting edge or working end) of each blade is placed flat on the stone; the blade will make a 45-degree angle with the stone. In the primary sharpening stroke, the cutting edge is moved forward. Unless the instrument is very dull, only two or three fairly heavy forward strokes will be necessary to sharpen the cutting edge.

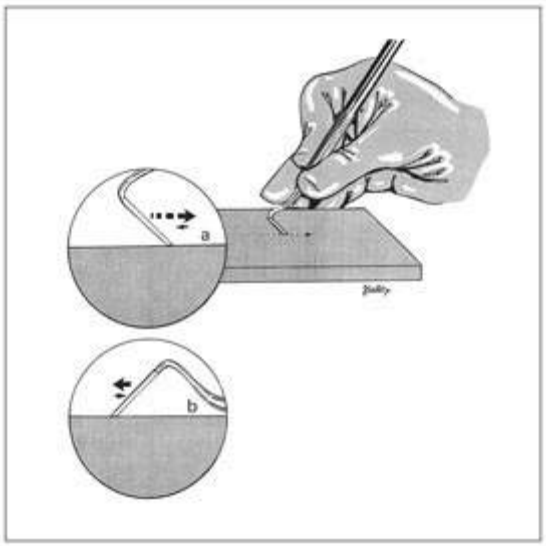


Fig 7-46 Sharpening the two ends of a double-ended binangle hoe: *(a)* sharpening the bevel end (outside or distal bevel); *(b)* sharpening the contrabevel end (inside or mesial bevel). The primary bevel is always flat against the stone; the face of the blade is up.

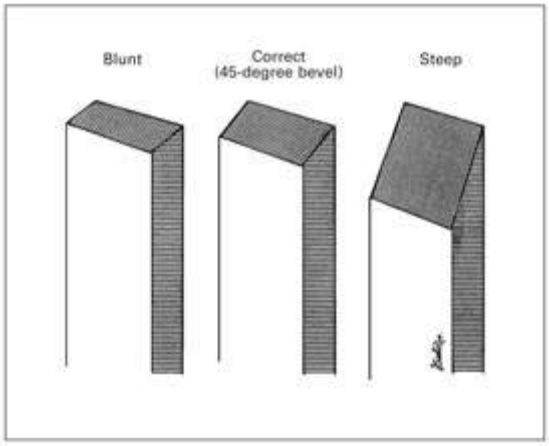


Fig 7-47 Bevels of sharpened cutting instruments. Working-end bevels of chisels, hatchets, and hoes, as well as the bevels of amalgam carvers, should be at approximately 45 degrees to the face of the blade. The cutting edge at the left is too blunt, the center blade has a correctly angled cutting edge, and the cutting edge at the right is too acute and will dull rapidly.

When spoons, discoid carvers, and cleoid carvers are sharpened, the instrument is rotated as the blade is advanced on the flat stone (Fig 7-48). The bevel is at 45 degrees, or slightly more or less, to the face, and the instrument is advanced on the stone with the bevel against the surface of the stone and the cutting edge of the instrument perpendicular to the path of advancement. When a blade with a rounded edge is being sharpened, the handle cannot simply be twirled to achieve the desired rotation but must actually be swung in an arc to keep the cutting edge of the blade perpendicular to the direction of the stroke and the bevel parallel with and against the surface of the stone.

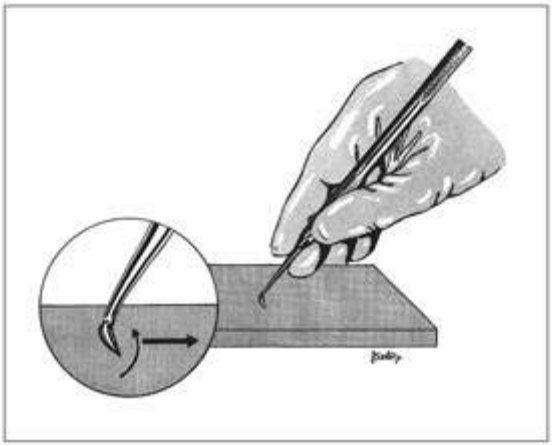


Fig 7-48 Sharpening a cleoid carver. The handle is swung in an arc to rotate the blade as the bevel is pulled forward on the stone. This movement is used to keep the cutting edge perpendicular to the direction of the stroke.

The discoid carver and spoon may be sharpened with a continuous rotation of the

blade; the shank moves clockwise from the 9 o'clock position to the 3 o'clock position in one motion. For the cleoid carver, however, the rotation begins with the shank in the 9 o'clock position and continues clockwise only until the bevel just next to the point is ground (see Fig 7-48); to sharpen the other side of the cleoid, the rotation begins with the shank at the 3 o'clock position and continues counterclockwise to the point. If sharpening both sides of a cleoid carver creates a sharp point at its tip, the junction of the bevels at the point should be slightly rounded.

The blade of a discoid spoon may be sharpened by grinding the face of the blade with a rotating stone (Fig 7-49). This method of sharpening also thins the blade, and care must be taken to avoid rendering the blade so thin that it could easily break.



Fig 7-49 Sharpening a discoid spoon with a rotating sharpening stone. A discoid spoon may also be sharpened on a flat stone; the blade is rotated as it is pulled with the cutting edge forward. If the face is ground with a rotating stone, the blade will be thinned and could be more likely to break during use.

Sharpening machines are available. A slowly rotating sharpening wheel is employed by one type of machine; an oscillating flat stone, or *hone*, is used by another. These machines are useful for sharpening instruments between patients and before sterilization.

When instruments are sharpened during an operative procedure, they should be sharpened with a sterile stone. When a stone is sterilized, it should not have oil in or on it, because the oil may thicken during the sterilization process and form a shellac-like coating that will prevent the abrasion needed for sharpening. A good substitute for oil is water. Stones lubricated with water should be washed well or cleaned in an ultrasonic cleaner after use to remove the metal filings prior to sterilization. A flat, white Arkansas stone or fine synthetic sharpening stone should be made a part of the operative dentistry instrument kit so that it is available during each procedure.

Mirrors, explorers, periodontal probes, and forceps

Mirrors, explorers, periodontal probes, and forceps are basic instruments that will

be needed during each appointment for diagnosis or treatment.

Mirrors. For every procedure performed in the mouth, the dentist must have clear and distinct vision of the field. Wherever possible, the field should be viewed with direct vision. When needed, the mouth mirror allows the operator to visualize areas of the mouth that he or she would not otherwise be able to see. It also allows the operator to maintain a body position that will reduce health problems associated with poor posture.

Almost as important as its allowing indirect visualization of obscure areas of the mouth is the mirror's function as a reflector of light into the area being examined or treated. A mirror that is positioned properly allows the operator to visualize the field of operation in the mirror and, at the same time, reflects the operating light into that area. To accomplish this, the light should be positioned behind and directed just to the side of the operator's head and into the mirror.

The mouth mirror can also serve as a retractor of soft tissue (tongue, cheeks, or lips) to aid access and visualization.

For clarity of vision, the reflective surface of the mirror should be on the external surface of the glass. This type of mirror is called a *front-surface mirror* (Fig 7-50a). Mouth mirrors are usually round and come in a variety of sizes (Fig 7-50b). The most widely used sizes for adults are the no. 4 and no. 5. For constricted areas in posterior regions of the mouth, when a rubber dam is in place, a smaller mirror, such as a no. 2, is helpful.

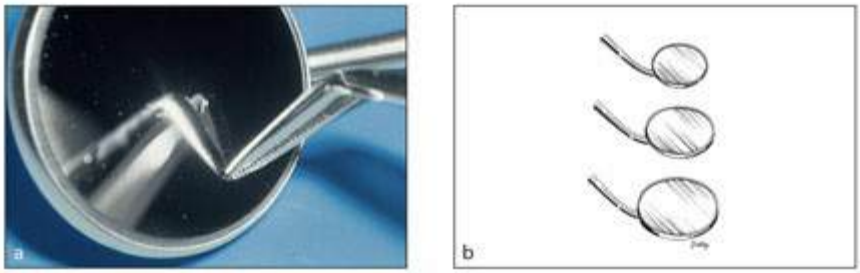


Fig 7-50*(a)* Front-surface mirror. Any object touching the mirror, such as the tips of the cotton forceps, will appear to be touching itself. *(b)* Mouth mirrors: *(top)* no. 2 (5/8-inch diameter); *(middle)* no. 4 (7/8-inch diameter); *(bottom)* no. 5 (15/16-inch diameter).

Explorers. *Explorers* are pointed instruments used to feel tooth surfaces for irregularities and to determine the hardness of exposed dentin. The explorer that is used most often is the shepherd's hook, or no. 23, explorer (Fig 7-51a). Another useful shape is a cowhorn explorer, which provides improved access for exploring proximal surfaces (Fig 7-51b). The no. 17 explorer is also useful in proximal areas

(Fig 7-51c).

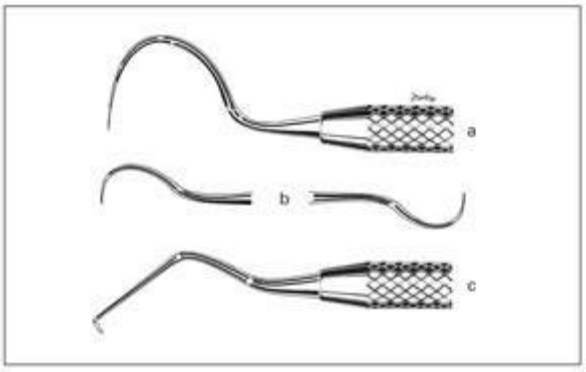


Fig 7-51 Dental explorers: *(a)* no. 23 explorer (shepherd's hook); *(b)* 3CH explorer (cowhorn or pigtail); *(c)* no. 17 explorer.

Periodontal probes. *Periodontal probes* are designed to detect and measure the depth of periodontal pockets. In operative dentistry, they are also used to determine dimensions of instruments and of various features of preparations or restorations. There are many periodontal probe designs; the differences are in the diameters, the position of the millimeter markings, the configuration of the markings (eg, whether they are notched or painted), and the design of the tip. Three commonly used probes are illustrated in Fig 7-52.

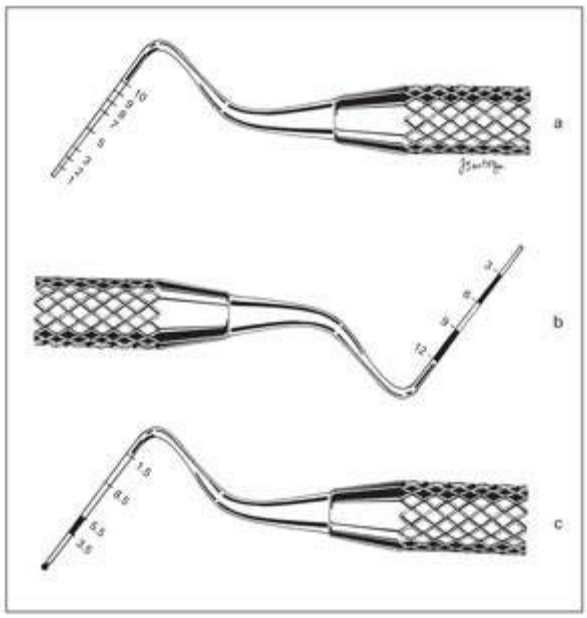


Fig 7-52 Periodontal probes: *(a)* QOW probe (Michigan O probe with Williams markings); *(b)* PCP12 probe (Marquis markings); *(c)* PSR (periodontal screening and recording) probe.

Forceps. Forceps of various kinds are useful in operative dentistry. *Cotton forceps* are used for picking up small items, such as cotton pellets (small cotton balls), and

carrying them to the mouth (Fig 7-53). Other forceps useful in operative dentistry include articulating paper forceps (Fig 7-54) and hemostatic forceps (hemostats) (Fig 7-55). A *hemostat* locks tightly, so it is often helpful in placing or removing items used to confine amalgam for condensation. *Articulating paper forceps* are designed to carry an inked tape to the mouth to mark the contacts of teeth in opposing arches during closure.

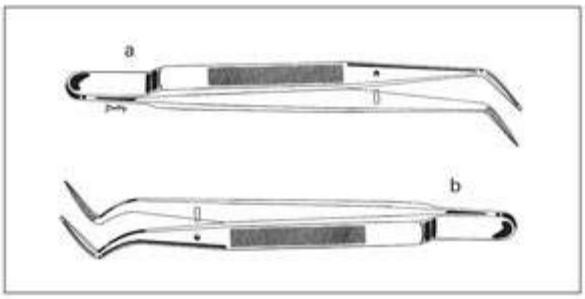


Fig 7-53 Cotton forceps: *(a)* College (no. 17); *(b)* Meriam (no. 18).

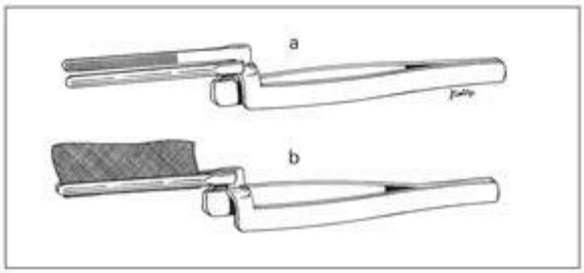


Fig 7-54 Articulating paper forceps: *(a)* Forceps handles provide a spring that keeps the jaws closed together; they are opened (as shown) by squeezing the handle. *(b)* The entire length of the piece of articulating paper or tape is supported by the jaws of the forceps.

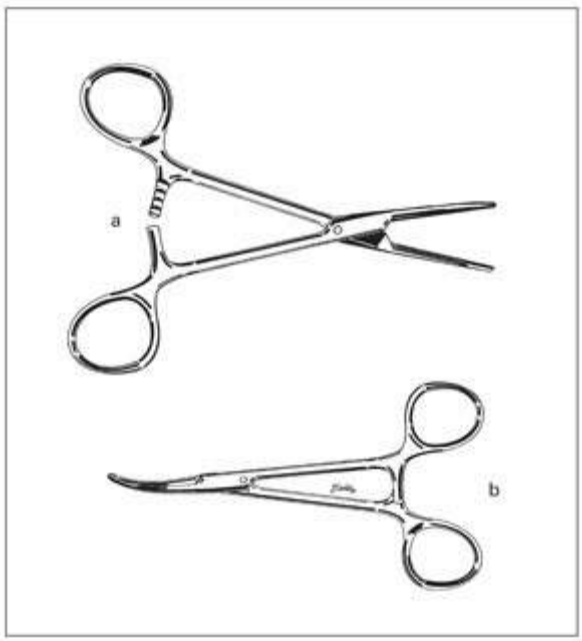


Fig 7-55 Hemostats: *(a)* Halstead mosquito straight, 7-inch; *(b)* Halstead mosquito curved, 5-inch.

Instrument grasps

The operator should master two basic instrument grasps, the *pen grasp*, which provides more flexibility of movement, and the *palm* or *palm-thumb grasp*, which provides limited movement but controlled power. Usually only one-handed grasps are used, but occasionally two-handed instrumentation is needed to make refinement of a preparation more precise (Fig 7-56).

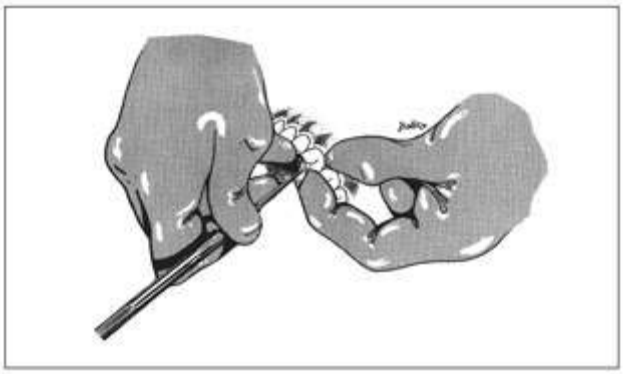


Fig 7-56 Two-handed instrumentation. The use of both hands can make refinement of a preparation more precise. The right hand is thrusting and rotating the instrument while the index finger of the left hand guides and assists the motion of the working end to refine a proximal margin of a Class 2 preparation. A similar dual-handed action is useful for condensing amalgam; it allows increased condensation force to be controlled.

Pen grasp. This is the most frequently used instrument grasp in operative dentistry. The pen grasp is actually different from the way one would grasp a pen (Fig 7-57); the handle of the instrument is engaged by the end, not the side, of the middle finger; this provides more finger power. The pen grasp is initiated by placement of the instrument handle between the thumb and index finger; the middle finger engages the handle near the shank or on the shank itself (Figs 7-57 and 7-58). The ring finger is braced against the teeth to stabilize instrument movement (see Fig 7-58).

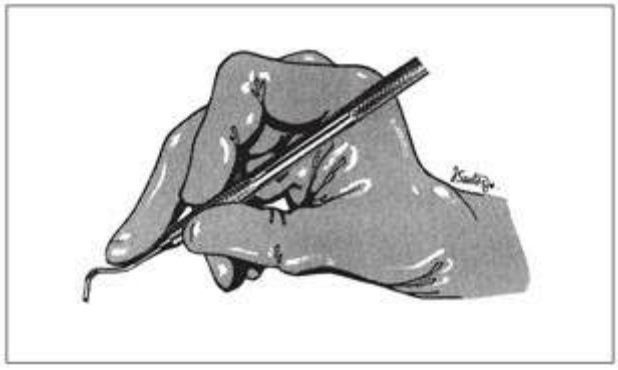


Fig 7-57 Pen grasp. The pen grasp is not actually the way a pen is held for writing. The instrument is held between the index finger and thumb, and the middle finger is placed atop the handle or shank, nearer the working end of the instrument, to provide more force, or thrust, directed toward the working end of the instrument.

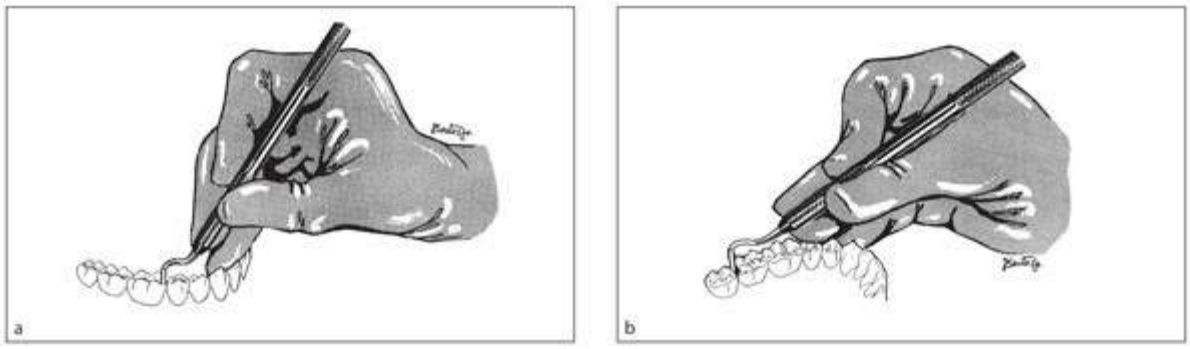


Fig 7-58*(a)* Pen grasp used in a chopping (downward) motion. The ring finger is resting on the incisal edges of the anterior teeth. During the use of any instrument in the mouth (with the exception of the mirror), a firm rest must be achieved on teeth or attached gingival tissue. *(b)* Pen grasp as the instrument is used more posteriorly and with a side-to-side or scraping motion. The small finger and ring finger are resting on the facial and occlusal surfaces, respectively.

Palm or palm-thumb grasp. In this grasp, the thumb serves as a brace (Fig 7-59). Side-to-side, rotating, or thrusting movements of the instrument by the wrist and fingers are controlled by the thumb, which is firmly in contact with the teeth (Fig 7-60).

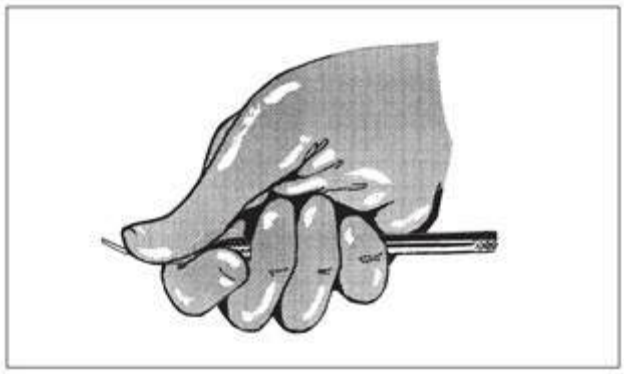


Fig 7-59 Palm-thumb grasp. The instrument is grasped much nearer to its end than in the pen grasp so that the thumb can be braced against the teeth to provide control during movement of the instrument.

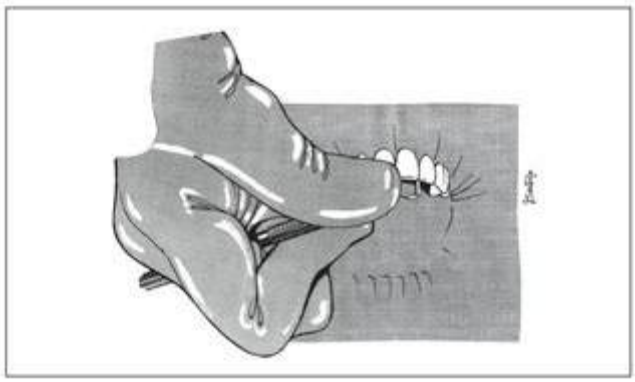


Fig 7-60 The palm-thumb grasp is used frequently when a hand cutting instrument, such as a gingival margin trimmer, is used in Class 3 preparations that have lingual access. The thumb is resting on the incisal edges of the teeth. The palm-thumb grasp is also used frequently with the Wedelstaedt chisel, usually for facial access in posterior and anterior operations, and occasionally for lingual access.

Instrument motions

The following are some of the many motions used with hand instruments:

- Chopping (in the direction of the working end of the instrument or parallel to the long axis of the blade)
- Pulling (toward the operator's hand)
- Pushing (away from the hand)
- Rotating
- Scraping (with the blade directed at an angle between 45 and 90 degrees to the surface being scraped and moved side to side or back and forth on the surface)
- Thrusting (forcibly pushing against a surface)

Rotating instruments

Handpieces

In dentistry, two basic types of handpieces are used: the straight handpiece (Fig 7-61) and the contra-angle handpiece (Fig 7-62). In the straight handpiece, the long axis of the bur is the same as the long axis of the handpiece. The straight handpiece is used more frequently for laboratory work but is occasionally useful clinically.



Fig 7-61 Straight handpiece. This handpiece is used occasionally in the mouth, but it is more frequently used extraorally, for tasks such as making adjustments to removable prostheses or adjusting and repolishing a cast gold or ceramic restoration prior to insertion. The bur installed in this handpiece is a tree-shaped denture bur.

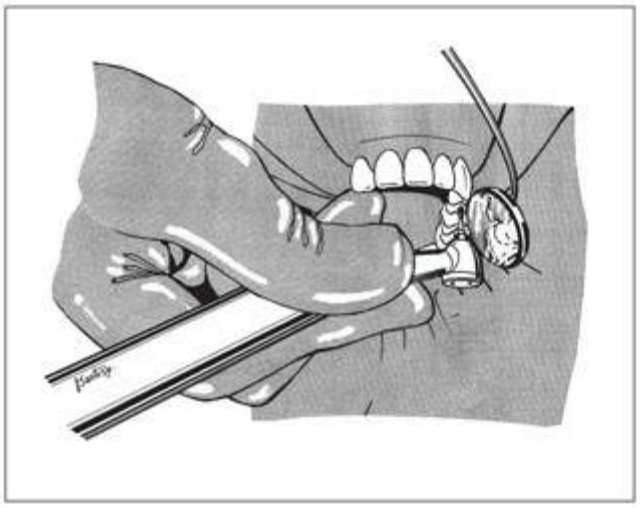


Fig 7-62 Contra-angle handpiece. This is a high-speed contra-angle handpiece, which is used with small-diameter burs for rapid cutting of tooth structure or restorations. A low-speed contra-angle is also useful for removal of carious dentin, with a slowly rotating round bur, and for shaping and polishing with abrasive disks and impregnated rubber polishers. Some operators also prefer the low-speed contra-angle for refining tooth preparations.

The primary handpiece used in the mouth is the contra-angle handpiece. As with hand instruments, *contra-angle* indicates that the head of the handpiece is angled first away from, and then back toward, the long axis of the handle. Also as with hand instruments, this contra-angle design is intended to bring the working point (the head of the bur) to within a few millimeters of the long axis of the handle of the handpiece to provide balance.

There are two types of contra-angle handpieces, which are classified by their

speed potential. Low-speed contra-angle handpieces have a typical free-running speed range of 500 to 15,000 rpm; some are able to slow to 200 rpm, and others are able to achieve speeds of 35,000 rpm. High-speed handpieces can achieve a free-running speed greater than 160,000 rpm, and some handpieces attain free-running speeds up to 500,000 rpm.[7] In the United States, most dentists are accustomed to air-turbine high-speed handpieces. The speed of these handpieces during tooth preparation is 180,000 rpm and lower, depending on the application pressure and the power of the handpiece. For air-turbine handpieces, speeds during tooth preparation are significantly less than their free-running speeds.

Electric handpieces (powered by an electric motor instead of an air-turbine) have been used for some time in Europe, and their use is rapidly growing in the United States. Most electric handpieces achieve free-running speeds of 200,000 rpm, an ideal speed for cutting enamel. Electric handpieces are very efficient in preparing teeth.

High-speed techniques are generally preferred for cutting enamel and dentin. Penetration through enamel and extension of the cavity outline are more efficient at high speed. Small-diameter burs should be used in the high-speed handpiece. High speed generates considerable heat, even with small-diameter burs, and should be used with air and water coolant sprays[8] and high-efficiency evacuation. For refining preparations, a high-speed handpiece may be slowed considerably and used with only air coolant and a gentle brushing or painting motion in which each application of the bur to the tooth is brief. This technique allows visualization and prevents overheating.[9]

Low-speed contra-angle handpieces, with round burs rotating very slowly, are used for removal of carious dentin. Low-speed contra-angle handpieces are also used for various finishing and polishing procedures that use abrasive disks, points, or cups.

There are two types of contra-angles based on their bur-locking, or *chucking*, mechanisms for the low-speed handpiece: a friction-grip chuck and a latch-type chuck. The shanks of the burs that fit into each of these types of contra-angle chuck are shown in Fig 7-63. The high-speed handpiece will receive only the friction-grip bur.

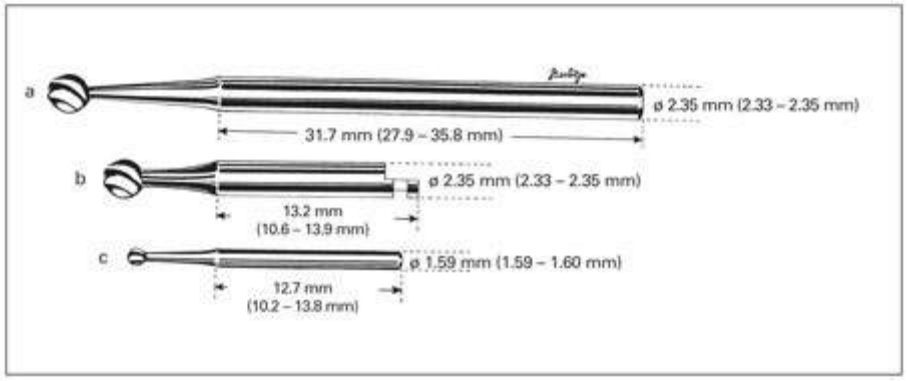


Fig 7-63 Typical dimensions (and ANSI/ADA standard dimension tolerances), in millimeters, of the three common bur designs: *(a)* straight handpiece bur; *(b)* latch-type bur for latch-type contra-angle handpiece; *(c)* friction-grip bur for friction-grip contra-angle handpiece.[10,11]

Burs

Hand-rotated dental instruments are known to have been used since the early 1700s. The foot engine came into use in dentistry in 1871 and the electric engine in 1872.4 The most significant advancement, which has made present-day high-speed cutting possible, is the tungsten carbide bur, which became available in 1947.[10]

Burs have three major parts: the head, the neck, and the shank (Fig 7-64). For the different types of handpieces or handpiece heads, there are burs with different designs and dimensions (see Fig 7-63).

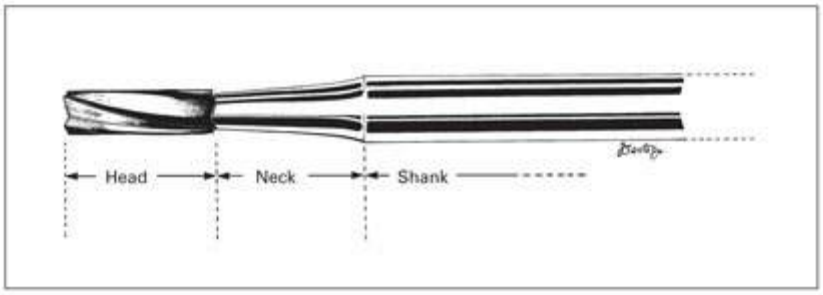


Fig 7-64 Parts of a rotary cutting instrument (bur).

The head of a bur is the portion that cuts. The cutting action is produced by blades on the head, and the blades are produced by cuts made into the head. The angle of the cutting edge of a blade (edge angle) is usually not acute; the angle is in the range of 90 degrees to provide strength to the blade and longevity of cutting efficiency of the bur. A cross section of a typical six-bladed bur is shown in Fig 7-65; the names of the faces and angles of the blades are also shown. The bur in Fig 7-65 has a negative rake angle, as do most burs used in dentistry.[10] The negative rake angle increases the life expectancy of the bur and provides for the most effective performance in low-

and high-speed ranges.

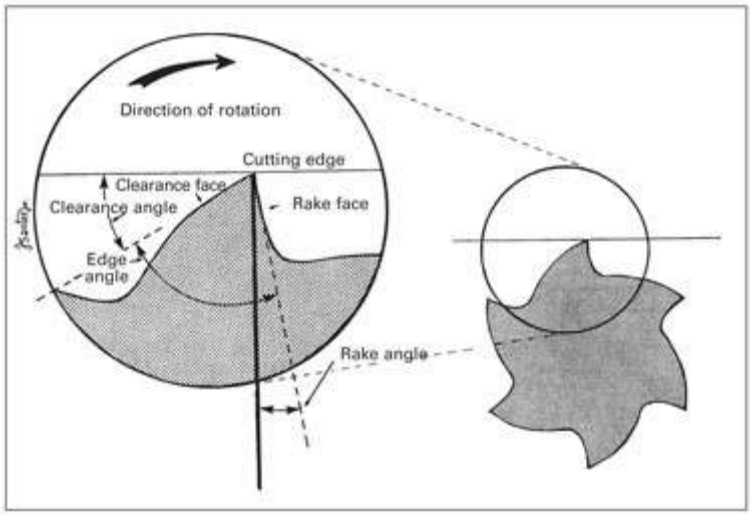


Fig 7-65 Typical bur head, viewed from the end of the bur nearest the handpiece.[5,10]

A positive rake angle would produce a more acute edge angle. Positive rake angles may be used to cut softer, weaker substances, such as soft carious dentin. If a blade with a positive rake angle were used to cut a hard material, such as sound enamel or dentin, it would dig in, leaving an irregularly cut surface, and the cutting edges of the blades would chip and dull rapidly.

The basic shapes of tooth-preparation burs used in operative dentistry are shown in Fig 7-66. Many other shapes are available; most are modifications of these five. Numbering systems have been introduced to describe the shapes of dental burs. The original system, introduced by SS White Dental Manufacturing, had nine shapes based on the burs available at that time.[10] That system has been modified and expanded as new burs have been developed. The American National Standards Institute/American Dental Association (ANSI/ADA) specification[11] provides standard characteristics for dental burs; this specification lists both the US numbers and the International Standards Organization (ISO) numbers for dental burs.

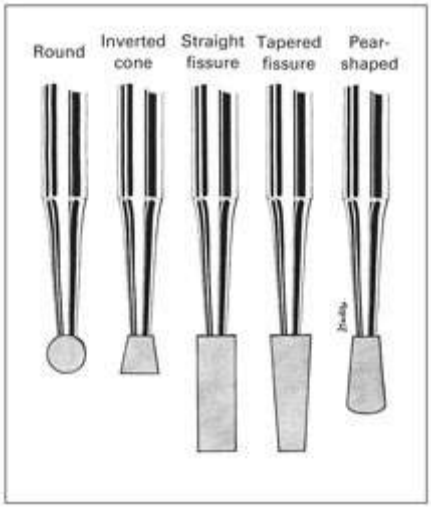


Fig 7-66 Basic bur head shapes for tooth preparation. Most burs used for tooth preparation are modifications of these burs. The primary modifications are lengthening of the bur heads and rounding of ends or corners to allow preparations to be cut without sharp line angles.

Prior to the advent of high-speed handpieces, it was found that additional cuts across the blades of a dental bur increased cutting efficiency; these cuts were called *crosscuts*. Today, with high-speed handpieces, crosscut burs are not normally of any benefit.

Table 7-1 shows diagrams, US bur sizes, and the head diameters of many available regular carbide tooth-preparation burs. ISO sizes for each type of bur can be calculated from the diameter: A bur with a diameter of 0.8 mm will have an ISO size of 008; a diameter of 1.0 mm will have an ISO size of 010. The ISO sizes are combined with the shape of the bur, so an ISO inverted cone 006 is an inverted cone bur with a 0.6-mm major diameter; from Table 7-1, it can be determined that an ISO inverted cone 006 corresponds with a US no. 33½ bur.

Table 7-1 Shapes and diameters of regular carbide burs used for tooth preparation (US designations*)



Round

Bur size	⅛₈	⅛	¼	½	1	2	3	4	5
Diameter (mm)	0.3	0.4	0.5	0.6	0.8	1.0	1.2	1.4	1.6

Bur size	6	7	8	9	11
Diameter (mm)	1.8	2.1	2.3	2.5	3.1



Inverted cone

Bur size	33½	34	35	36	37	39	40
Diameter (mm)	0.6	0.8	1.0	1.2	1.4	1.8	2.1



Straight fissure†

Bur size	55½	56	57	58	59	60
Diameter (mm)	0.6	0.8	1.0	1.2	1.4	1.6



Straight fissure, rounded end (straight dome)†

Bur size	1156	1157	1158
Diameter (mm)	0.8	1.0	1.2



Straight fissure, crosscut†

Bur size	556	557	558	559	560
Diameter (mm)	0.8	1.0	1.2	1.4	1.6



Straight fissure, rounded end, crosscut (straight dome crosscut)

Bur size	1556	1557	1558
Diameter (mm)	0.8	1.0	1.2



Tapered fissure†

Bur size	168	169	170	171
Diameter (mm)	0.8	0.9	1.0	1.2



Tapered fissure, rounded end (tapered dome)†

Bur size	1169	1170	1171
Diameter (mm)	0.9	1.0	1.2



Tapered fissure, crosscut†

Bur size	699	700	701	702	703
Diameter (mm)	0.9	1.0	1.2	1.6	2.1



Pear†

Bur size	329	330	331	332
Diameter (mm)	0.6	0.8	1.0	1.2



Long inverted cone, rounded corners (amalgam preparation)

Bur size	245	246
Diameter (mm)	0.8	1.2



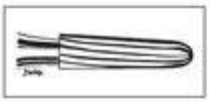
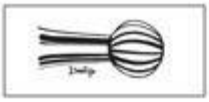
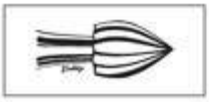
End-cutting

Bur size	956	957
Diameter (mm)	0.8	1.0

* Adapted from American National Standards Institute/American Dental Association Specification 23 and catalogs of Midwest Dental Products and Brasseler.

† Some sizes available with a long head (L).

Table 7-2 Shapes and diameters of some of the available 12-bladed carbide finishing burs used for smooth cuts in tooth preparation and for finishing restorations (US designations*)

	**Egg**							
	Bur size	7404	7406	7408				
	Diameter (mm)	1.4	1.8	2.3				
	**Bullet**							
	Bur size	7801	7802	7803				
	Diameter (mm)	0.9	1.0	1.2				
	**Needle**							
	Bur size	7901	7902	7903				
	Diameter (mm)	0.9	1.0	1.2				
	**Round**							
	Bur size	7002	7003	7004	7006	7008	7009	7010
	Diameter (mm)	1.0	1.2	1.4	1.8	2.3	2.7	3.1
	**Flame**							
	Bur size	7102	7104	7106	7108			
	Diameter (mm)	1.2	1.4	1.8	2.3			
	**Cone**							
	Bur size	7202	7204	7205	7206			
	Diameter (mm)	1.0	1.4	1.6	1.8			
	**Long pear (inverted taper)**							
	Bur size	7302	7303	7304				
	Diameter (mm)	1.0	1.2	1.4				
	**Straight fissure**							
	Bur size	7572	7583					
	Diameter (mm)	1.0	1.2					
	**Taper**							
	Bur size	7702	7713					
	Diameter (mm)	1.0	1.2					

*Adapted from catalogs of Midwest Dental Products and Brasseler.

It is useful to know the diameters and lengths of the burs used in tooth preparation so that they can be used as gauges of depth and distance. Bur head lengths may vary from manufacturer to manufacturer, so it is best to measure the individual burs being used and to use their dimensions as references for measuring preparation dimensions.

Another type of bur that is very useful in operative dentistry is the trimming and finishing bur. These burs come in a variety of shapes and sizes; the heads of some trimming and finishing burs are shown in Table 7-2. Trimming and finishing burs are excellent for making very smooth cuts in tooth preparations, for adjusting occlusion in enamel or of a restoration, and for contouring and finishing restorations. Trimming and finishing burs have more blades than tooth-preparation burs, and the greater the number of blades, the smoother the cut surface that can be attained. The number of blades necessary for a desired surface smoothness varies with the diameter of the

bur; typical trimming and finishing burs have 8 to 12, 16 to 20, or 30 blades.

Many designs of trimming and finishing burs are available for shaping and contouring esthetic restorations. One extremely useful type of trimming and finishing bur that is available with 8, 16, or 30 blades is a straight-sided taper with a safe (noncutting) end (called *esthetic trimming*, or *ET*, *burs* from Brasseler). This bur is designed so that the end will rest on the tooth surface without cutting tooth structure and allow contouring of the adjacent restoration; it is available in several different lengths and diameters, as well as with the different numbers of blades.

Diamonds. Used increasingly in operative dentistry, diamond burs are especially useful for preparations for bonded restorations. Several manufacturers produce diamonds that mimic the shapes of many of the carbide burs. Diamond burs cut tooth structure well and are acceptable substitutes for carbide burs, but many of the smaller sizes are not available as diamond burs. Diamond burs with fine-grit diamond surfaces are also useful for contouring and polishing esthetic restorations.

Air-abrasion technology

In the 1940s, an instrument called the Airdent (SS White) was introduced as a means of cavity preparation.[12] Because all restorations placed at that time depended on cavity-preparation shape for retention, and because the Airdent did not prepare undercuts in preparations, the technology soon lost favor. When it was reintroduced in the 1980s, it received a greater degree of acceptance because bonded restorations had become routine.[13] Etched enamel and dentin, rather than the shape of the cavity preparation, give retention to many restorations. A large number of air-abrasion units are being marketed for opening fissures, for some cavity preparations, and to facilitate repair of existing restorations with bonding technology. An example of a compact and relatively inexpensive air abrasion unit is pictured in Fig 7-67a. More recently, technology has been introduced that includes a curtain of water around the spray of alumina particles (Fig 7-67b). This water curtain reduces the amount of dust generated in the procedure.

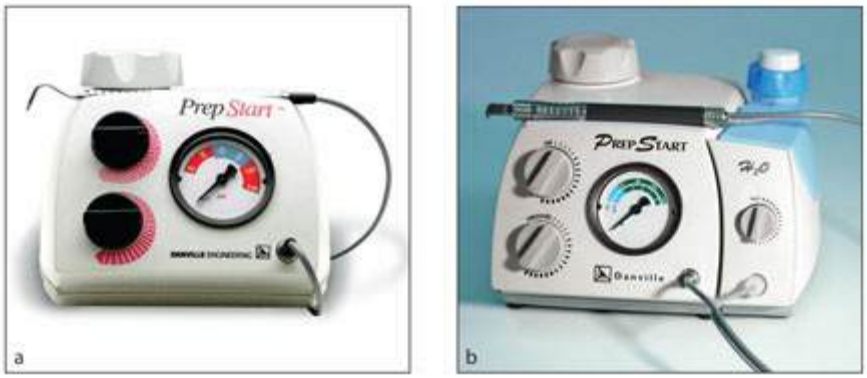


Fig 7-67 *(a)* PrepStart air-abrasion cavity-preparation unit (Danville Materials). *(b)* PrepStart H2O air-abrasion cavity-preparation unit (Danville Materials). This unit is an example of the use of new technology that provides a water curtain around the stream of alumina particles. This greatly reduces the amount of dust that is produced.

Sonic and ultrasonic technology

The use of sonic and ultrasonic technology for tooth preparation has been referenced in the literature for over two decades, especially in conjunction with endodontic biomechanical preparation,[14] and these tools remain popular for endodontic indications.[15] In addition, ultrasonic and sonic tooth preparation is a viable means for conservative cavity preparation.[16] Additional information on ultrasonic cavity preparation is provided in chapter 11.

Visualization and magnification

The quality, and therefore the serviceability and longevity, of dental restorations is dependent on the ability of the operator to see what he or she is doing. One of the primary advantages of the rubber dam in operative dentistry is improvement of the visualization of the operating field. Most current contra-angle handpieces have fiber-optic systems by which lights are placed in the contra-angle heads to improve visualization of the operating field.

Magnification devices are extremely helpful in restorative procedures, and some form of magnification is recommended for every dentist providing restorative dentistry services.[17] Available magnification devices run the gamut of effectiveness and expense. Among the finest magnifiers are the telescopes (Fig 7-68), which are the most expensive. Less expensive loupes are available from several manufacturers (Fig 7-69).



Fig 7-68*(a and b)* Binocular telescopes (in-the-lens type) manufactured by Designs for Vision. *(c)* Binocular telescopes (in-the-lens type) manufactured by Orascoptic. *(d)* Binocular telescopes (frame-mounted flip-down type) manufactured by Orascoptic. *(e)* Binocular telescopes (headband-mounted flip-down type) manufactured by Orascop-tic.

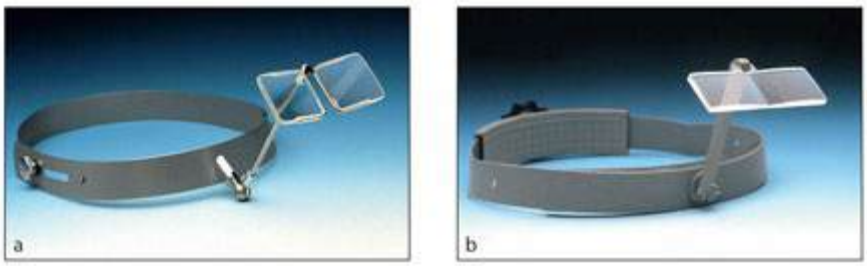


Fig 7-69*(a)* Binocular loupes manufactured by Almore International. *(b)* Binocular loupes manufactured by Edroy Products.

In choosing a magnification device, the operator is wise to select one that gives a focal distance in the range of 10 to 14 inches. The 2.0- to 4.0-diopter range is recommended. In addition, magnifiers are available that are mounted into the lenses of eyeglasses (see Figs 7-68a to 7-68c) or that flip down either from the glasses frame (see Fig 7-68d) or a headband (see Figs 7-68e and 7-69).

References

1. Glossary of Operative Dentistry Terms, ed 1. Washington, DC: Academy of Operative Dentistry, 1983.
2. Black GV. A Work on Operative Dentistry. Chicago: Medico-Dental, 1908.
3. Metals Handbook, ed 10, vol 1. Materials Park, OH: American Society of Metals, 1990:841–842,908–909.

4. Ring ME. Dentistry: An Illustrated History. New York: Abrams, 1985:251.
5. Charbeneau GT. Principles and Practice of Operative Dentistry, ed 3. Philadelphia: Lea & Febiger, 1988.
6. Webster's Ninth New Collegiate Dictionary. Springfield, MA: Merriam-Webster, 1988.
7. Young JM. Dental air-powered handpieces: Selection, use, and sterilization. Compend Contin Educ Dent 1993;14:358–366.
8. Lauer H, Kraft E, Rothlauf W, Zwingers T. Effects of the temperature of cooling water during high-speed and ultrahigh-speed tooth preparation. J Prosthet Dent 1990;63:407–414.
9. Bouschor CR, Matthews JL. A four-year clinical study of teeth restored after preparation with an air turbine handpiece with an air coolant. J Prosthet Dent 1966;16:306–309.
10. Roberson TM, Heymann HO, Swift EJ. Sturdevant's Art and Science of Operative Dentistry, ed 5. St Louis: Mosby-Elsevier, 2006.
11. American National Standards Institute/American Dental Association Specification No. 23 (Revised) for Dental Excavating Burs. Chicago: American Dental Association, 1982.
12. Black RB. Technique for nonmechanical preparation of cavities and prophylaxis. J Am Dent Assoc 1945;32:955–965.
13. Goldstein RE, Parkins FM. Air-abrasive technology: Its new role in restorative dentistry. J Am Dent Assoc 1994;125:551–557.
14. Dummer PM, Alodeh MH, Doller R. Shaping of simulated root canals in resin blocks using files activated by a sonic handpiece. Int Endod J 1989;22:211–215.
15. Kanter V, Weldon E, Nair U, et al. A quantitative and qualitative analysis of ultrasonic versus sonic endodontic systems on canal cleanliness and obturation. Oral Surg Oral Med Oral Pathol Oral Radiol Endod 2011;112:809–813.
16. Koubi S, Tassery H. Minimally invasive dentistry using sonic and ultrasonic devices in ultraconservative Class 2 restorations. J Contemp Dent Pract 2008;9:155–165.
17. Christensen GJ. Magnification. Clin Res Assoc Newsletter 1990;14(10):1.